Sustainable Forest Management

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Summary

Most respondents indicated that deforestation, charcoal production, and fire frequency were decreasing, wildlife populations were increasing, and natural resource management was improving within CFs. Most FDG groups also indicated that both deforestation and charcoal production were decreasing outside their CF, suggesting there is no leakage effect. Community female FDGs tended to have a more positive view of the impact of CFM compared to the other groups. However, it is important to note that a small number of FDG groups and individual interviewees suggested the situation was worsening in their CFs. It will be important to identify CFs that are doing less well and address the problems. Few CFMGs were practising natural forest restoration or establishing plantations. Hence, there is a considerable opportunity to expand the contribution of CFM to meeting national restoration and plantation targets.

Introduction

In this chapter, we consider the effect of community forestry on indicators of sustainable forest management. The information is based only on the responses of interviewees to our survey tool. A separate chapter will analyse satellite derived remote sensed data on fire frequency and deforestation.

Deforestation

We asked interviewees whether CFM has reduced deforestation inside their CF and whether or not deforestation has increased outside their CF. An increase in deforestation outside the community forest might represent leakage, whereby activities involving clearing forest that are proscribed in the CF are displaced to areas around the CF.

Deforestation inside the community forest

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    deforestation = case_match(
            impact_on_sfm_cfm_reduced_forest_clearing_cfma,
            "no_there_has_never_been_clearing_for_agriculture" ~ "No change",
            "Agricultural practices ceased way before CFM" ~ "No change",
            "One field for cassava" ~ "No change",
            "Yes there no more fields_  it's all forests" ~ "Significant",
            "Yes_ significant reduction. There was chitemene system before CFM" ~ "Significan
            "no_increase" ~ "No change",
            "yes_moderate" ~ "Moderate",
            "yes_significant" ~ "Significant",
            .default = NA
    ),
    deforestation = factor(deforestation, levels = c("Significant","Moderate","No change")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(deforestation) |>
  count(interview_type, deforestation) |>
  group_by(interview_type) |>
  mutate(
    prop = n / sum(n)
  ) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = deforestation)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
```

```r
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c(
    "No change"         = "#C0392B",
    "Moderate" = "#F0C040",
    "Significant"        = "#2A9D8F"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(
  y    = NULL,
  fill = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

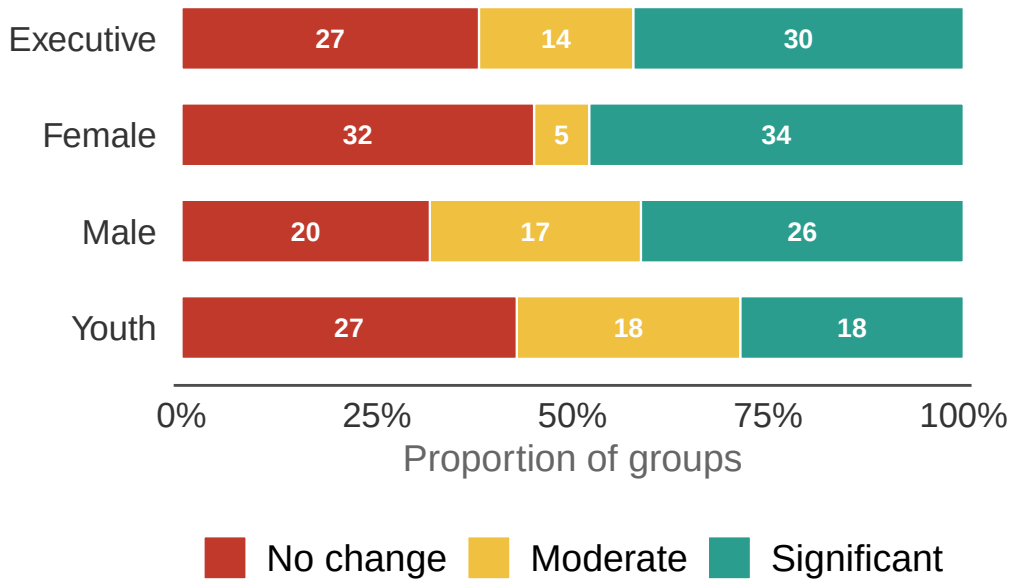

Figure 1: 'Has CFM reduced deforestation inside the community forest?' The length of each bar segment represents the proportion of FGD groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

A majority of groups report either a moderate or significant reduction in deforestation inside the CF. Moreover, these answers were consistent across the FGD groups. There were no significant differences in response.

> **Deforestation - FGDs - Model methods**
>
> To examine the differences among FGD types in their assessment of whether CFM had reduced deforestation, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_cfm_reduced_forest_clearing_cfma,
      "no_there_has_never_been_clearing_for_agriculture" ~ "No change",
      "Agricultural practices ceased way before CFM"     ~ "No change",
      "One field for cassava"                            ~ "No change",
```

```r
      "Yes there no more fields_  it's all forests"      ~ "Significant",
      "Yes_ significant reduction. There was chitemene system before CFM" ~ "Significant",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("No change", "Moderate", "Significant"),
                           ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(deforestation) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M1 <- clmm(deforestation ~
              interview_type +
              (1 | province) +
              (1 | province_cfmg),
            data = test.data)

emm.M1     <- emmeans(M1, specs = ~ interview_type)
pairwise.M1 <- pairs(emm.M1, adjust = "Tukey")
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: deforestation ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  268  -268.27 550.54 314(1251) 1.34e-05 6.5e+01

Random effects:
```

```
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3587   0.5989
 province      (Intercept) 1.1242   1.0603
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale -0.09229    0.34968  -0.264    0.792
interview_typeMale    0.17689    0.34780   0.509    0.611
interview_typeYouth  -0.56897    0.34710  -1.639    0.101

Threshold coefficients:
                     Estimate Std. Error z value
No change|Moderate    -0.5714     0.4609  -1.240
Moderate|Significant   0.4295     0.4601   0.933


 contrast           estimate    SE  df z.ratio p.value
 Executive - Female   0.0923 0.350 Inf   0.264  0.9936
 Executive - Male    -0.1769 0.348 Inf  -0.509  0.9571
 Executive - Youth    0.5690 0.347 Inf   1.639  0.3565
 Female - Male       -0.2692 0.358 Inf  -0.752  0.8758
 Female - Youth       0.4767 0.354 Inf   1.346  0.5334
 Male - Youth         0.7459 0.357 Inf   2.091  0.1560

P value adjustment: tukey method for comparing a family of 4 estimates
```

Among the individual interviewees (Table 1), again a majority indicated that CFM had reduced deforestation inside the CF. The differences among the interviewee roles were not significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_cfm_reduced_forest_clearing_cfma,
      "no_there_has_never_been_clearing_for_agriculture" ~ "No change",
      "Agricultural practices ceased way before CFM"     ~ "No change",
      "One field for cassava"                            ~ "No change",
      "Yes there no more fields_  it's all forests"      ~ "Significant",
      "Yes_ significant reduction. There was chitemene system before CFM" ~ "Significant",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
```

```r
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("No change", "Moderate", "Significant")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(deforestation)) |>
  count(interviewee_role, deforestation, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, deforestation, prop) |>
  pivot_wider(names_from = deforestation, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(`No change`, Moderate, Significant), decimals = 0) |>
  tab_header(
    title    = "Perceived reduction in deforestation inside the community forest by intervie
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 1: Perceived reduction in deforestation inside the community forest by individual interviewee role.

Perceived reduction in deforestation inside the community forest by interviewee role
Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	No change	Moderate	Significant
CFMG executive member	60	17%	23%	60%
Community member	29	28%	24%	48%
Traditional leader	70	50%	17%	33%
District Forestry Officer	51	22%	51%	27%
NGO representative	30	20%	40%	40%

Reduced forest clearing - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had reduced deforestation, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_cfm_reduced_forest_clearing_cfma,
      "no_there_has_never_been_clearing_for_agriculture" ~ "No change",
      "Agricultural practices ceased way before CFM"     ~ "No change",
      "One field for cassava"                            ~ "No change",
      "Yes there no more fields_  it's all forests"      ~ "Significant",
      "Yes_ significant reduction. There was chitemene system before CFM" ~ "Significant",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("No change", "Moderate", "Significant"),
                           ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
```

```r
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(deforestation) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M2 <- clmm(deforestation ~
              interviewee_role +
              interviewee_gender +
              interviewee_age +
              (1 | province) +
              (1 | province_cfmg),
            data = test.data)

emm.M2     <- emmeans(M2, specs = ~ interviewee_role)
pairwise.M2 <- pairs(emm.M2, adjust = "Tukey")
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula:
deforestation ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  240  -243.93 507.86 510(1531) 1.86e-03 7.2e+04

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3452   0.5875
 province      (Intercept) 0.3046   0.5519
Number of groups:  province_cfmg 72,  province 8

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.07465    0.51196   0.146   0.8841
interviewee_roletrad_leader      -0.94320    0.48982  -1.926   0.0542 .
interviewee_roledfo              -0.70076    0.49304  -1.421   0.1552
interviewee_roleNGO_rep          -0.22466    0.54517  -0.412   0.6803
interviewee_gendermale            0.77552    0.36163   2.144   0.0320 *
interviewee_age                  -0.01590    0.01207  -1.317   0.1878
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Threshold coefficients:
                      Estimate Std. Error z value
No change|Moderate   -1.525308   0.803968  -1.897
Moderate|Significant -0.008527   0.790196  -0.011


 contrast                        estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member  -0.0747 0.512 Inf  -0.146  0.9999
 cfmg_member - trad_leader        0.9432 0.490 Inf   1.926  0.3036
 cfmg_member - dfo                0.7008 0.493 Inf   1.421  0.6138
 cfmg_member - NGO_rep            0.2247 0.545 Inf   0.412  0.9940
 cfmg_exec_member - trad_leader   1.0179 0.429 Inf   2.375  0.1222
 cfmg_exec_member - dfo           0.7754 0.413 Inf   1.875  0.3307
 cfmg_exec_member - NGO_rep       0.2993 0.490 Inf   0.611  0.9735
 trad_leader - dfo               -0.2424 0.410 Inf  -0.592  0.9763
 trad_leader - NGO_rep           -0.7185 0.487 Inf  -1.475  0.5787
 dfo - NGO_rep                   -0.4761 0.444 Inf  -1.073  0.8204

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Deforestation outside the community forest

To investigate possible leakage effects, we also asked interviewees whether CFM had led to an increase in deforestation outside their community forest.

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_forest_clearing_increased_outside_cfma,
      "Clearing only done for field opening" ~ "No change",
      "In Mboshya there are no agricultural activities" ~ "No change",
      "It increases according to seasons_ especially during the tobacco harvesting" ~ "Modera
      "No change_ community members' concentration is on farming. They have appreciation for
      "There has never been  cutting of trees out side the cfma" ~ "No change",
      "There has never been any forest clearing outside the cfma." ~ "No change",
      "There has never been cutting of trees" ~ "No change",
      "no_decrease" ~ "Decrease",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
```

```r
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("Significant", "Moderate", "No change", "Decrease")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(deforestation) |>
  count(interview_type, deforestation) |>
  group_by(interview_type) |>
  mutate(
    prop = n / sum(n)
  ) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = deforestation)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = 0.01),
    name   = "Proportion of groups"
  ) +
  scale_fill_manual(
    values = c(
      "Decrease" = "#2A9D8F",
      "No change"   = "#76B7A0",
      "Moderate"    = "#F0C040",
      "Significant" = "#C0392B"
    ),
    guide = guide_legend(reverse = TRUE)
```

```r
  ) +
  labs(
    y    = NULL,
    fill = NULL
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1),    colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

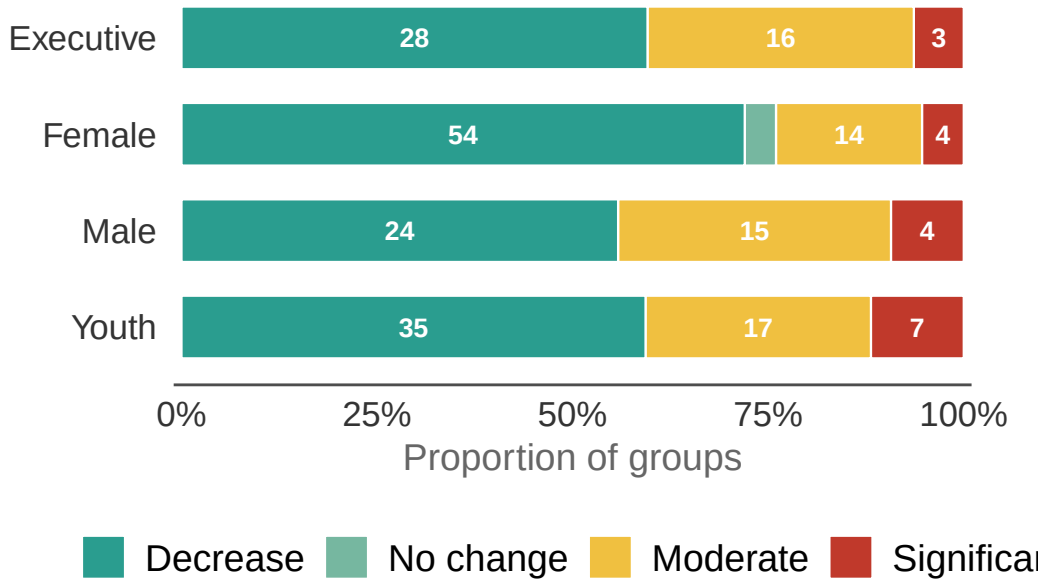

Figure 2: 'Has CFM increased deforestation outside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Contrary to the expectations of leakage, a majority of the FGD groups responded that CFM actually reduced deforestation outside their CF. Moreover, the responses were consistent across FGD groups. There were no significant differences in response.

> **Deforestation outside the CF - FGDs - Model methods**
>
> To examine the differences among FGD types in their assessment of whether CFM had increased deforestation outside the CF, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_forest_clearing_increased_outside_cfma,
      "Clearing only done for field opening" ~ "No change",
      "In Mboshya there are no agricultural activities" ~ "No change",
      "It increases according to seasons_ especially during the tobacco harvesting" ~ "Modera
      "No change_ community members' concentration is on farming. They have appreciation for
      "There has never been  cutting of trees out side the cfma" ~ "No change",
      "There has never been any forest clearing outside the cfma." ~ "No change",
      "There has never been cutting of trees" ~ "No change",
      "no_decrease" ~ "Decrease",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("Significant", "Moderate", "No change", "Decrease")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
```

```r
  ) |>
  drop_na(deforestation) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M3 <- clmm(deforestation ~
              interview_type +
              (1 | province) +
              (1 | province_cfmg),
            data = test.data)

emm.M3     <- emmeans(M3, specs = ~ interview_type)
pairwise.M3 <- pairs(emm.M3, adjust = "Tukey")
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: deforestation ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  224  -194.95 405.90 358(1077) 9.15e-05 9.7e+02

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.2955   0.5436
 province      (Intercept) 0.4744   0.6887
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale   0.7928     0.4216   1.881    0.060 .
interview_typeMale    -0.1685     0.4469  -0.377    0.706
interview_typeYouth    0.1451     0.4235   0.343    0.732
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
Significant|Moderate  -2.5936     0.4813  -5.388
Moderate|No change    -0.5117     0.4143  -1.235
No change|Decrease    -0.4427     0.4137  -1.070

 contrast           estimate    SE  df z.ratio p.value
```

```
 Executive - Female   -0.793 0.422 Inf  -1.881  0.2364
 Executive - Male      0.168 0.447 Inf   0.377  0.9817
 Executive - Youth    -0.145 0.423 Inf  -0.343  0.9862
 Female - Male         0.961 0.424 Inf   2.266  0.1061
 Female - Youth        0.648 0.393 Inf   1.648  0.3515
 Male - Youth         -0.314 0.430 Inf  -0.730  0.8852

P value adjustment: tukey method for comparing a family of 4 estimates
```

Among the individual interviewees (Table 2), the individuals from the communities agreed that deforestation outside their CF had decreased. However, NGO representatives and DFOs were less likely to suggest deforestation had decreased outside the CF and more likely to suggest a moderate increase in deforestation.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_forest_clearing_increased_outside_cfma,
      "Clearing only done for field opening" ~ "No change",
      "In Mboshya there are no agricultural activities" ~ "No change",
      "It increases according to seasons_ especially during the tobacco harvesting" ~ "Modera
      "No change_ community members' concentration is on farming. They have appreciation for
      "There has never been  cutting of trees out side the cfma" ~ "No change",
      "There has never been any forest clearing outside the cfma." ~ "No change",
      "There has never been cutting of trees" ~ "No change",
      "no_decrease" ~ "Decrease",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("Decrease", "No change", "Moderate", "Significant")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
```

```r
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(deforestation)) |>
  count(interviewee_role, deforestation, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, deforestation, prop) |>
  pivot_wider(names_from = deforestation, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(Decrease, `No change`, Moderate, Significant), decimals = 0) |>
  tab_header(
    title    = "Perceived increase in deforestation outside the community forest by intervie
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 2: Perceived increase in deforestation outside the community forest by individual interviewee role.

Perceived increase in deforestation outside the community forest by interviewee role
Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	Moderate	Significant	No change
CFMG executive member	56	75%	16%	9%	0%
Community member	28	50%	39%	11%	0%
Traditional leader	68	72%	21%	6%	1%
District Forestry Officer	40	25%	52%	20%	2%

NGO representative	23	39%	48%	13%	0%

Deforestation outside the CF - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had increased deforestation outside the community forest, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    deforestation = case_match(
      impact_on_sfm_forest_clearing_increased_outside_cfma,
      "Clearing only done for field opening" ~ "No change",
      "In Mboshya there are no agricultural activities" ~ "No change",
      "It increases according to seasons_ especially during the tobacco harvesting" ~ "Modera
      "No change_ community members' concentration is on farming. They have appreciation for
      "There has never been  cutting of trees out side the cfma" ~ "No change",
      "There has never been any forest clearing outside the cfma." ~ "No change",
      "There has never been cutting of trees" ~ "No change",
      "no_decrease" ~ "Decrease",
      "no_increase"    ~ "No change",
      "yes_moderate"   ~ "Moderate",
      "yes_significant" ~ "Significant",
      .default = NA
    ),
    deforestation = factor(deforestation,
                           levels = c("Significant", "Moderate", "No change", "Decrease")),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(deforestation) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M4 <- clmm(deforestation ~
              interviewee_role +
              interviewee_gender +
              interviewee_age +
              (1 | province) +
```

```r
              (1 | province_cfmg),
            data = test.data)

emm.M4     <- emmeans(M4, specs = ~ interviewee_role)
pairwise.M4 <- pairs(emm.M4, adjust = "Tukey")
```

The differences in opinion between the CFMG executive members and NGO representatives and DFOs, and between traditional leaders and NGO representatives and DFOs, were significant.

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula:
deforestation ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  215  -183.40 388.80 719(2314) 2.09e-04 8.8e+04

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.0000   0.0000
 province      (Intercept) 0.5163   0.7185
Number of groups:  province_cfmg 70,  province 8

Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  1.19569    0.51842   2.306   0.0211 *
interviewee_roletrad_leader       1.13187    0.50198   2.255   0.0241 *
interviewee_roledfo              -0.83252    0.50769  -1.640   0.1010
interviewee_roleNGO_rep          -0.39550    0.58203  -0.680   0.4968
interviewee_gendermale           -0.27144    0.39812  -0.682   0.4954
interviewee_age                  -0.01402    0.01281  -1.094   0.2738
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
Significant|Moderate  -2.9332     0.8944  -3.279
Moderate|No change    -0.7579     0.8681  -0.873
No change|Decrease    -0.7059     0.8679  -0.813
```

```
 contrast                        estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member  -1.1957 0.518 Inf  -2.306  0.1427
 cfmg_member - trad_leader       -1.1319 0.502 Inf  -2.255  0.1598
 cfmg_member - dfo                0.8325 0.508 Inf   1.640  0.4718
 cfmg_member - NGO_rep            0.3955 0.582 Inf   0.680  0.9609
 cfmg_exec_member - trad_leader   0.0638 0.476 Inf   0.134  0.9999
 cfmg_exec_member - dfo           2.0282 0.467 Inf   4.347  0.0001
 cfmg_exec_member - NGO_rep       1.5912 0.572 Inf   2.783  0.0429
 trad_leader - dfo                1.9644 0.482 Inf   4.077  0.0004
 trad_leader - NGO_rep            1.5274 0.569 Inf   2.685  0.0562
 dfo - NGO_rep                   -0.4370 0.523 Inf  -0.836  0.9194

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Charcoal production

We asked interviewees whether CFM has reduced charcoal production inside their CF, and whether it has increased immediately outside their CF.

Charcoal production inside the community forest

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_in_cfma,
      "No. There have never been  charcoal  production  in this forest"    ~ "No change",
      "No_ charcoal production is not a common practice for community members"    ~ "No chang
      "Some areas are encroached by charcoal production"    ~ "No change",
      "There has never been  charcoal  production"    ~ "No change",
      "There has never been  charcoal  production in this forest"    ~ "No change",
      "yes_reduced"   ~ "Moderate",
      "yes_high" ~ "Significant",
      "no" ~ "Significant",
      "no_stopped" ~ "Significant",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("Significant", "Moderate", "No change")),
    interview_type = factor(
```

```r
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(charcoal) |>
count(interview_type, charcoal) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = charcoal)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c(
    "No change"   = "#C0392B",
    "Moderate"    = "#F0C040",
    "Significant" = "#2A9D8F"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
```

```
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1),    colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

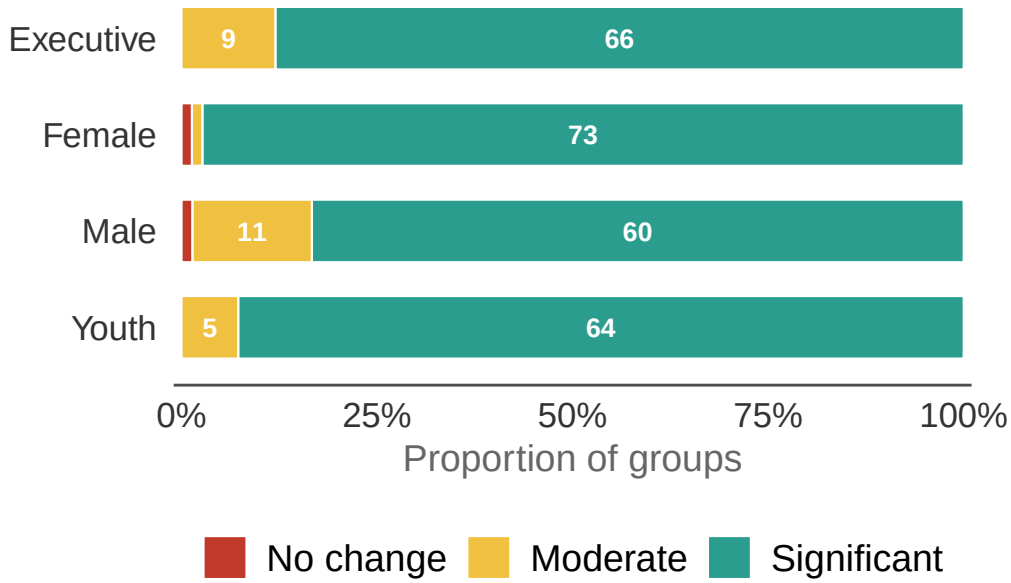

Figure 3: 'Has CFM reduced charcoal production inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Almost all the FGD groups agreed that charcoal making inside their CF had decreased, with most indicating that charcoal making had decreased significantly. The small differences in responses among FGD groups were not significant.

Charcoal production inside the CF - FGDs - Model methods

To examine the differences among FGD types in their assessment of whether CFM had reduced charcoal production inside the CF, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD

types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_in_cfma,
      "No. There have never been  charcoal  production  in this forest"    ~ "No change",
      "No_ charcoal production is not a common practice for community members"    ~ "No chang
      "Some areas are encroached by charcoal production"    ~ "No change",
      "There has never been  charcoal  production"    ~ "No change",
      "There has never been  charcoal  production in this forest"    ~ "No change",
      "yes_reduced"   ~ "Moderate",
      "yes_high" ~ "Significant",
      "no" ~ "Significant",
      "no_stopped" ~ "Significant",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("No change", "Moderate", "Significant"),
                      ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(charcoal) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$charcoal) >= 2) {
  M5 <- clmm(charcoal ~
               interview_type +
               (1 | province) +
               (1 | province_cfmg),
             data = test.data)
```

```r
  emm.M5     <- emmeans(M5, specs = ~ interview_type)
  pairwise.M5 <- pairs(emm.M5, adjust = "Tukey")
} else {
  M5 <- NULL
  cat("Model not fitted: only", nlevels(test.data$charcoal),
      "level(s) present in charcoal after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: charcoal ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik AIC    niter     max.grad cond.H
 logit flexible  291  -91.66 197.31 258(1028) 1.13e-05 2.4e+01

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 1.0542   1.0267
 province      (Intercept) 0.2734   0.5229
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale   1.7745     0.8403   2.112   0.0347 *
interview_typeMale    -0.3462     0.5185  -0.668   0.5044
interview_typeYouth    0.5419     0.6157   0.880   0.3787
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
No change|Moderate    -5.3963     0.9235  -5.844
Moderate|Significant  -2.4428     0.5450  -4.482


 contrast           estimate    SE  df z.ratio p.value
 Executive - Female   -1.775 0.840 Inf  -2.112  0.1493
 Executive - Male      0.346 0.519 Inf   0.668  0.9093
 Executive - Youth    -0.542 0.616 Inf  -0.880  0.8151
 Female - Male         2.121 0.818 Inf   2.594  0.0468
 Female - Youth        1.233 0.892 Inf   1.383  0.5103
```

```
 Male - Youth         -0.888 0.600 Inf  -1.480  0.4496

P value adjustment: tukey method for comparing a family of 4 estimates
```

Likewise, among the individual interviewees (Table 3), almost all interviewees agreed that there had been a substantial decrease in charcoal making inside their CF. Again, this result was consistent across interviewee roles (there were no significant differences among roles).

```
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_in_cfma,
      "No. There have never been  charcoal  production  in this forest"    ~ "No change",
      "No_ charcoal production is not a common practice for community members"    ~ "No chang
      "Some areas are encroached by charcoal production"    ~ "No change",
      "There has never been  charcoal  production"    ~ "No change",
      "There has never been  charcoal  production in this forest"    ~ "No change",
      "yes_reduced"   ~ "Moderate",
      "yes_high" ~ "Significant",
      "no" ~ "Significant",
      "no_stopped" ~ "Significant",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("No change", "Moderate", "Significant")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(charcoal)) |>
  count(interviewee_role, charcoal, name = "obs") |>
  group_by(interviewee_role) |>
```

```
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, charcoal, prop) |>
  pivot_wider(names_from = charcoal, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(`No change`, Moderate, Significant), decimals = 0) |>
  tab_header(
    title    = "Perceived reduction in charcoal production inside the community forest by int
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 3: Perceived reduction in charcoal production inside the community forest by individual interviewee role.

Perceived reduction in charcoal production inside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Moderate	Significant	No change
CFMG executive member	63	14%	86%	0%
Community member	31	26%	74%	0%
Traditional leader	74	11%	86%	3%
District Forestry Officer	50	20%	80%	0%
NGO representative	30	17%	80%	3%

Charcoal production inside the CF - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had reduced charcoal production inside the CF, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We

included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_in_cfma,
      "No. There have never been  charcoal  production  in this forest"    ~ "No change",
      "No_ charcoal production is not a common practice for community members"    ~ "No chang
      "Some areas are encroached by charcoal production"    ~ "No change",
      "There has never been  charcoal  production"    ~ "No change",
      "There has never been  charcoal  production in this forest"    ~ "No change",
      "yes_reduced"   ~ "Moderate",
      "yes_high" ~ "Significant",
      "no" ~ "Significant",
      "no_stopped" ~ "Significant",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("No change", "Moderate", "Significant"),
                      ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(charcoal) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$charcoal) >= 2) {
  M6 <- clmm(charcoal ~
               interviewee_role +
               interviewee_gender +
               interviewee_age +
               (1 | province) +
               (1 | province_cfmg),
             data = test.data)
  emm.M6     <- emmeans(M6, specs = ~ interviewee_role)
  pairwise.M6 <- pairs(emm.M6, adjust = "Tukey")
} else {
  M6 <- NULL
  cat("Model not fitted: only", nlevels(test.data$charcoal),
```

```
      "level(s) present in charcoal after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: charcoal ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  248  -117.84 255.68 474(1886) 1.40e-04 6.9e+04

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3035   0.5509
 province      (Intercept) 0.6238   0.7898
Number of groups:  province_cfmg 74,  province 8

Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.829165   0.635386   1.305    0.192
interviewee_roletrad_leader       0.421898   0.636456   0.663    0.507
interviewee_roledfo               0.006291   0.641719   0.010    0.992
interviewee_roleNGO_rep           0.255626   0.711879   0.359    0.720
interviewee_gendermale           -0.891869   0.604163  -1.476    0.140
interviewee_age                  -0.001294   0.016146  -0.080    0.936

Threshold coefficients:
                     Estimate Std. Error z value
No change|Moderate     -5.443      1.238  -4.398
Moderate|Significant   -2.424      1.088  -2.227


 contrast                       estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member -0.82917 0.635 Inf  -1.305  0.6881
 cfmg_member - trad_leader      -0.42190 0.636 Inf  -0.663  0.9642
 cfmg_member - dfo              -0.00629 0.642 Inf  -0.010  1.0000
 cfmg_member - NGO_rep          -0.25563 0.712 Inf  -0.359  0.9965
 cfmg_exec_member - trad_leader  0.40727 0.616 Inf   0.662  0.9645
 cfmg_exec_member - dfo          0.82287 0.594 Inf   1.385  0.6376
 cfmg_exec_member - NGO_rep      0.57354 0.675 Inf   0.850  0.9148
 trad_leader - dfo               0.41561 0.595 Inf   0.698  0.9570
```

```
 trad_leader - NGO_rep           0.16627 0.720 Inf   0.231  0.9994
 dfo - NGO_rep                  -0.24933 0.650 Inf  -0.384  0.9954

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Charcoal production outside the community forest

We also asked interviewees whether CFM has led to an increase in charcoal production outside their CF.

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_increased_outside_cfma,
      "no_reduced"    ~ "Decrease",
      "no_change"    ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      "Yes_ moderate increase. Only for household use"   ~ "Moderate",
      "Moderate decrease"    ~ "Decrease",
      "Due to increase in population_ some charcoal production now being observed in some ope
      "CFMG members not involved in charcoal production"    ~ "No change",
      "Charcoal  production  is not allowed in the chiefdom"    ~ "No change",
      "Charcoal  production is not allowed in the chiefdom"    ~ "No change",
      "Charcoal production  is not allowed  in the chiefdom"    ~ "No change",
      "Charcoal production has been banned from the chiefdom"    ~ "No change",
      "Charcoal production is not common in this part of the district"    ~ "No change",
      "Charcoal production is not commonly practiced. There is no market for charcoal product
      "Charcoal production is not practiced in the community"    ~ "No change",
      "Charcoal production not a common practice in the community"    ~ "No change",
      "Charcoal production only done when clearing an area for new agricultural fields"    ~
      "No charcoal  production"    ~ "No change",
      "No charcoal  production In this area"    ~ "No change",
      "No charcoal  production in this community"    ~ "No change",
      "No charcoal production"    ~ "No change",
      "No charcoal production near CFMA"    ~ "No change",
      "No. There has never been charcoal  production"    ~ "No change",
      .default = NA
    ),
    charcoal = factor(charcoal,
```

```r
                      levels = c("Significant", "Moderate", "No change", "Decrease")),
  interview_type = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(charcoal) |>
count(interview_type, charcoal) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = charcoal)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c(
    "Decrease"    = "#2A9D8F",
    "No change"   = "#76B7A0",
    "Moderate"    = "#F0C040",
    "Significant" = "#C0392B"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
```

```
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1),    colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

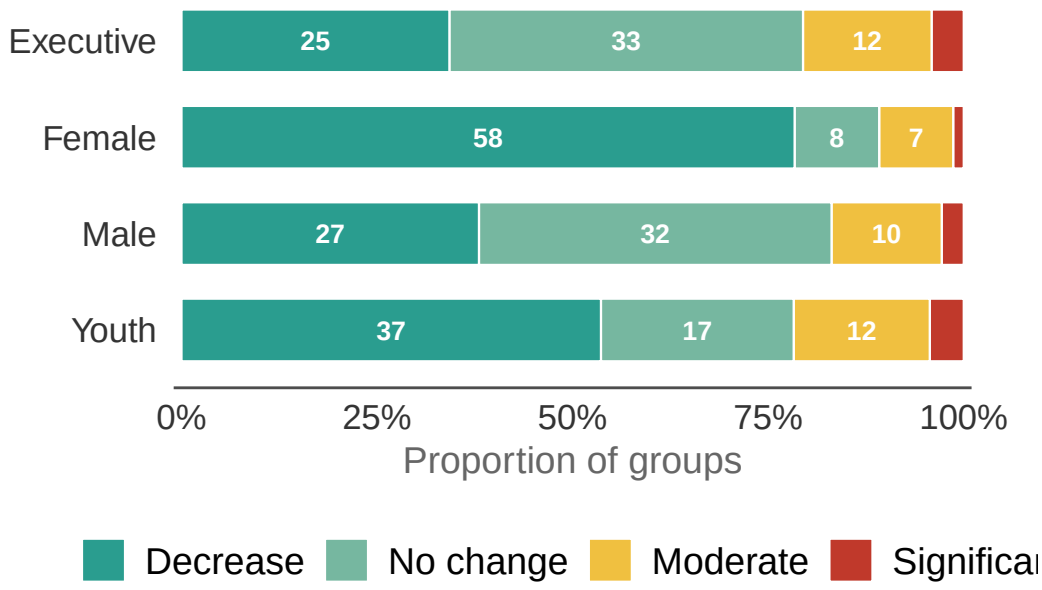

Figure 4: 'Has CFM increased charcoal production outside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen, the vast majority of groups indicated that charcoal making had also decreased outside their CF. However, there were a small number that indicated charcoal making had increased, some indicating it had increased significantly. Female FGD groups were significantly more likely to indicate that charcoal making had decreased outside the CF than all other groups (CFMG executive members, male or youth community members). There were no significant

differences among these other groups, who most often indicated that there was no change in the amount of charcoal making.

> **Charcoal production outside the CF - FGDs - Model methods**
>
> To examine the differences among FGD types in their assessment of whether CFM had increased charcoal production outside the community forest, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_increased_outside_cfma,
      "no_reduced"    ~ "Decrease",
      "no_change"    ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      "Yes_ moderate increase. Only for household use"   ~ "Moderate",
      "Moderate decrease"    ~ "Decrease",
      "Due to increase in population_ some charcoal production now being observed in some ope
      "CFMG members not involved in charcoal production"    ~ "No change",
      "Charcoal  production  is not allowed in the chiefdom"    ~ "No change",
      "Charcoal  production is not allowed in the chiefdom"    ~ "No change",
      "Charcoal production  is not allowed  in the chiefdom"    ~ "No change",
      "Charcoal production has been banned from the chiefdom"    ~ "No change",
      "Charcoal production is not common in this part of the district"    ~ "No change",
      "Charcoal production is not commonly practiced. There is no market for charcoal product
      "Charcoal production is not practiced in the community"    ~ "No change",
      "Charcoal production not a common practice in the community"    ~ "No change",
      "Charcoal production only done when clearing an area for new agricultural fields"    ~
      "No charcoal  production"    ~ "No change",
      "No charcoal  production In this area"    ~ "No change",
      "No charcoal  production in this community"    ~ "No change",
      "No charcoal production"    ~ "No change",
      "No charcoal production near CFMA"    ~ "No change",
      "No. There has never been charcoal  production"    ~ "No change",
      .default = NA
    ),
    charcoal = factor(charcoal,
```

```r
                        levels = c("Decrease", "No change", "Moderate", "Significant"),
                        ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(charcoal) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$charcoal) >= 2) {
  M7 <- clmm(charcoal ~
               interview_type +
               (1 | province) +
               (1 | province_cfmg),
             data = test.data)
  emm.M7     <- emmeans(M7, specs = ~ interview_type)
  pairwise.M7 <- pairs(emm.M7, adjust = "Tukey")
} else {
  M7 <- NULL
  cat("Model not fitted: only", nlevels(test.data$charcoal),
      "level(s) present in charcoal after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: charcoal ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  287  -290.88 597.77 371(1487) 4.49e-04 4.6e+01

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3769   0.6139
```

```
 province      (Intercept) 0.8389   0.9159
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale  -1.9042     0.3834  -4.967 6.81e-07 ***
interview_typeMale    -0.1563     0.3161  -0.494   0.6210
interview_typeYouth   -0.5805     0.3341  -1.738   0.0823 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
Decrease|No change    -0.3697     0.4166  -0.888
No change|Moderate     1.4936     0.4342   3.440
Moderate|Significant   3.4731     0.5393   6.440


 contrast           estimate    SE  df z.ratio p.value
 Executive - Female    1.904 0.383 Inf   4.967  <.0001
 Executive - Male      0.156 0.316 Inf   0.494  0.9604
 Executive - Youth     0.580 0.334 Inf   1.738  0.3041
 Female - Male        -1.748 0.386 Inf  -4.528  <.0001
 Female - Youth       -1.324 0.394 Inf  -3.358  0.0044
 Male - Youth          0.424 0.337 Inf   1.259  0.5893

P value adjustment: tukey method for comparing a family of 4 estimates
```

Among the individual interviewees (Table 4), similar to the results about deforestation, most community interviewees suggested there had been a decrease or no change in charcoal production outside the CF. However, NGO representatives and DFOs were significantly more likely to indicate that there had been an increase in charcoal production outside the CF.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_increased_outside_cfma,
      "no_reduced"    ~ "Decrease",
      "no_change"    ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      "Yes_ moderate increase. Only for household use"   ~ "Moderate",
```

```
      "Moderate decrease"    ~ "Decrease",
      "Due to increase in population_ some charcoal production now being observed in some ope
      "CFMG members not involved in charcoal production"    ~ "No change",
      "Charcoal  production  is not allowed in the chiefdom"    ~ "No change",
      "Charcoal  production is not allowed in the chiefdom"    ~ "No change",
      "Charcoal production  is not allowed  in the chiefdom"    ~ "No change",
      "Charcoal production has been banned from the chiefdom"    ~ "No change",
      "Charcoal production is not common in this part of the district"    ~ "No change",
      "Charcoal production is not commonly practiced. There is no market for charcoal product
      "Charcoal production is not practiced in the community"    ~ "No change",
      "Charcoal production not a common practice in the community"    ~ "No change",
      "Charcoal production only done when clearing an area for new agricultural fields"    ~
      "No charcoal  production"    ~ "No change",
      "No charcoal  production In this area"    ~ "No change",
      "No charcoal  production in this community"    ~ "No change",
      "No charcoal production"    ~ "No change",
      "No charcoal production near CFMA"    ~ "No change",
      "No. There has never been charcoal  production"    ~ "No change",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("Decrease", "No change", "Moderate", "Significant")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(charcoal)) |>
  count(interviewee_role, charcoal, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, charcoal, prop) |>
  pivot_wider(names_from = charcoal, values_from = prop, values_fill = 0) |>
```

```
arrange(interviewee_role) |>
gt() |>
cols_label(interviewee_role = "Role") |>
fmt_percent(columns = c(Decrease, `No change`, Moderate, Significant), decimals = 0) |>
tab_header(
  title    = "Perceived increase in charcoal production outside the community forest by int
  subtitle = "Values represent the % of respondents in each response category. n = the tota
) |>
opt_align_table_header(align = "left") |>
cols_align(align = "center", columns = -interviewee_role) |>
cols_align(align = "left", columns = interviewee_role) |>
tab_style(
  style     = cell_text(align = "center"),
  locations = cells_column_labels(columns = interviewee_role)
) |>
sub_missing()
```

Table 4: Perceived increase in charcoal production outside the community forest by individual interviewee role.

Perceived increase in charcoal production outside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	No change	Moderate	Significant
CFMG executive member	63	73%	8%	13%	6%
Community member	30	73%	7%	20%	0%
Traditional leader	73	70%	14%	10%	7%
District Forestry Officer	45	40%	20%	27%	13%
NGO representative	30	50%	17%	30%	3%

Charcoal production outside the CF - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had increased charcoal production outside the community forest, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```r
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    charcoal = case_match(
      impact_on_sfm_charcoal_production_increased_outside_cfma,
      "no_reduced"    ~ "Decrease",
      "no_change"    ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      "Yes_ moderate increase. Only for household use"   ~ "Moderate",
      "Moderate decrease"    ~ "Decrease",
      "Due to increase in population_ some charcoal production now being observed in some ope
      "CFMG members not involved in charcoal production"    ~ "No change",
      "Charcoal  production  is not allowed in the chiefdom"    ~ "No change",
      "Charcoal  production is not allowed in the chiefdom"    ~ "No change",
      "Charcoal production  is not allowed  in the chiefdom"    ~ "No change",
      "Charcoal production has been banned from the chiefdom"    ~ "No change",
      "Charcoal production is not common in this part of the district"    ~ "No change",
      "Charcoal production is not commonly practiced. There is no market for charcoal product
      "Charcoal production is not practiced in the community"    ~ "No change",
      "Charcoal production not a common practice in the community"    ~ "No change",
      "Charcoal production only done when clearing an area for new agricultural fields"    ~
      "No charcoal  production"    ~ "No change",
      "No charcoal  production In this area"    ~ "No change",
      "No charcoal  production in this community"    ~ "No change",
      "No charcoal production"    ~ "No change",
      "No charcoal production near CFMA"    ~ "No change",
      "No. There has never been charcoal  production"    ~ "No change",
      .default = NA
    ),
    charcoal = factor(charcoal,
                      levels = c("Decrease", "No change", "Moderate", "Significant"),
                      ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(charcoal) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$charcoal) >= 2) {
  M8 <- clmm(charcoal ~
```

```r
                interviewee_role +
                interviewee_gender +
                interviewee_age +
                (1 | province) +
                (1 | province_cfmg),
              data = test.data)
  emm.M8     <- emmeans(M8, specs = ~ interviewee_role)
  pairwise.M8 <- pairs(emm.M8, adjust = "Tukey")
} else {
  M8 <- NULL
  cat("Model not fitted: only", nlevels(test.data$charcoal),
      "level(s) present in charcoal after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: charcoal ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  241  -229.45 480.91 606(1897) 1.64e-04 1.3e+05

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.4324   0.6576
 province      (Intercept) 0.5048   0.7105
Number of groups:  province_cfmg 74,  province 8

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.14217    0.59812   0.238  0.81212
interviewee_roletrad_leader      -0.08832    0.57276  -0.154  0.87745
interviewee_roledfo               1.84748    0.61566   3.001  0.00269 **
interviewee_roleNGO_rep           1.56261    0.65084   2.401  0.01635 *
interviewee_gendermale            0.09286    0.38642   0.240  0.81010
interviewee_age                   0.04292    0.01352   3.176  0.00150 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
```

```
Decrease|No change     3.2872     0.9921   3.314
No change|Moderate     4.0987     1.0119   4.051
Moderate|Significant   5.8754     1.0714   5.484

 contrast                       estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member  -0.1422 0.598 Inf  -0.238  0.9993
 cfmg_member - trad_leader        0.0883 0.573 Inf   0.154  0.9999
 cfmg_member - dfo               -1.8475 0.616 Inf  -3.001  0.0226
 cfmg_member - NGO_rep           -1.5626 0.651 Inf  -2.401  0.1150
 cfmg_exec_member - trad_leader   0.2305 0.473 Inf   0.487  0.9886
 cfmg_exec_member - dfo          -1.7053 0.476 Inf  -3.582  0.0031
 cfmg_exec_member - NGO_rep      -1.4204 0.530 Inf  -2.681  0.0567
 trad_leader - dfo               -1.9358 0.497 Inf  -3.895  0.0009
 trad_leader - NGO_rep           -1.6509 0.551 Inf  -2.997  0.0228
 dfo - NGO_rep                    0.2849 0.477 Inf   0.597  0.9756

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Fire

We asked interviewees about fire monitoring practices and whether CFM had reduced fire incidence inside their CF. We did not ask about changes in fire frequency outside the CF, because clearing fields for agriculture or improving grazing often involves burning.

Fire monitoring

```r
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    response = substr(impact_on_sfm_fire_monitoring_in_cfma, 1, 1),
    response = case_match(
      response,
      "y" ~ "Yes",
      "n" ~ "No"
    )
  ) |>
  filter(!is.na(response)) |>
  count(province, response) |>
  mutate(
    n_signed = if_else(response == "No", -n, n),
```

```r
  response  = factor(response, levels = c("No", "Yes"))
) |>
ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name   = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide  = guide_legend(reverse = FALSE)
) +
labs(
  y    = NULL,
  fill = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

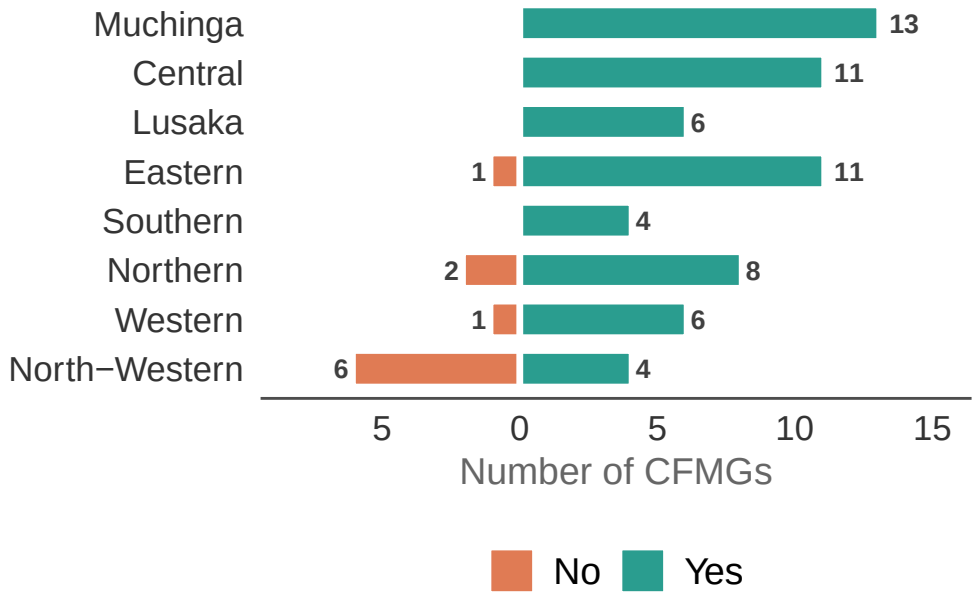

Figure 5: Whether CFMGs have established fire monitoring in the community forest by province. The length of the bars represents the number of CFMGs either monitoring fires (Yes) or not (No). The numbers on the bars represent the actual number of CFMGs in each category.

As can be seen in the figure, most CFMGs have instigated a system of fire monitoring. However, ten CFMGs across four provinces from our sample of 75 did not have any system of fire monitoring.

Fire incidence inside the community forest

We asked interviewees whether CFM has reduced fire incidence inside their CF.

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    fire = case_match(
      impact_on_sfm_fire_incidence_reduction_cfma,
      "no_incidents_have_increased" ~ "Increase",
      "no_change"     ~ "No change",
      "yes_moderate_reduction"    ~ "Moderate",
      "yes_significant_reduction" ~ "Significant",
```

```r
      "No change observed as at now" ~ "No change",
      .default = NA
    ),
    fire = factor(fire, levels = c("Significant", "Moderate", "No change","Increase")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(fire) |>
  count(interview_type, fire) |>
  group_by(interview_type) |>
  mutate(
    prop = n / sum(n)
  ) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = fire)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = 0.01),
    name   = "Proportion of groups"
  ) +
  scale_fill_manual(
    values = c(
      "Increase" = "#C0392B",
      "No change"   = "#F0C040",
      "Moderate"    = "#2A9D8F",
      "Significant" = "#76B7A0"
    ),
    guide = guide_legend(reverse = TRUE)
```

```r
  ) +
  labs(
    y    = NULL,
    fill = NULL
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

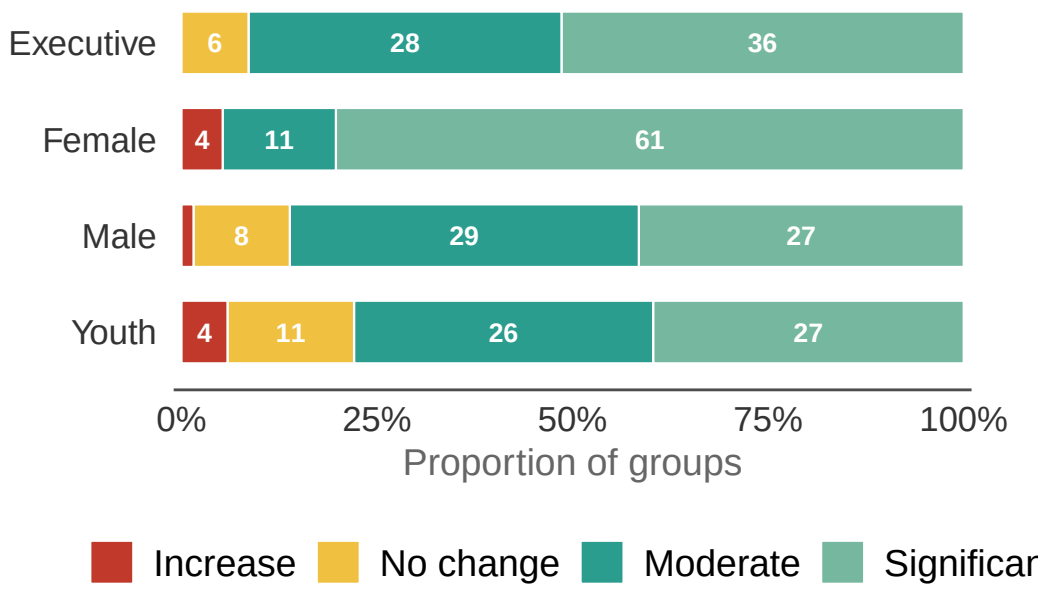

Figure 6: 'Has CFM reduced fire incidence inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen, a majority of groups reported a moderate or significant reduction in fire incidence in the CF. Female FDGs reported more significant declines in fire frequency compared to the other groups, who reported more moderate declines. These differences between female FDG groups and the other FDG groups (CFMG Executives, male and youth) were significant.

> **Fire incidence - FGDs - Model methods**
>
> To examine the differences among FGD types in their assessment of whether CFM had reduced fire incidence, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```r
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    fire = case_match(
      impact_on_sfm_fire_incidence_reduction_cfma,
      "no_incidents_have_increased" ~ "Increase",
      "no_change"     ~ "No change",
      "yes_moderate_reduction"    ~ "Moderate",
      "yes_significant_reduction" ~ "Significant",
      "No change observed as at now" ~ "No change",
      .default = NA
    ),
    fire = factor(fire,
                  levels = c("Increase","No change", "Moderate", "Significant"),
                  ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(fire) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()
```

```r
if (nlevels(test.data$fire) >= 2) {
  M9 <- clmm(fire ~
               interview_type +
               (1 | province) +
               (1 | province_cfmg),
             data = test.data)
  emm.M9      <- emmeans(M9, specs = ~ interview_type)
  pairwise.M9 <- pairs(emm.M9, adjust = "Tukey")
} else {
  M9 <- NULL
  cat("Model not fitted: only", nlevels(test.data$fire),
      "level(s) present in fire after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: fire ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  279  -256.76 529.52 372(1119) 2.52e-05 3.9e+01

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.7007   0.8371
 province      (Intercept) 1.0854   1.0418
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale   1.5672     0.4172   3.757 0.000172 ***
interview_typeMale    -0.4754     0.3542  -1.342 0.179480
interview_typeYouth   -0.7028     0.3506  -2.005 0.044997 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
Increase|No change    -4.2643     0.6101  -6.989
No change|Moderate    -2.6140     0.5118  -5.108
Moderate|Significant  -0.1858     0.4662  -0.399
```

```
 contrast           estimate    SE  df z.ratio p.value
 Executive - Female   -1.567 0.417 Inf  -3.757  0.0010
 Executive - Male      0.475 0.354 Inf   1.342  0.5358
 Executive - Youth     0.703 0.351 Inf   2.005  0.1862
 Female - Male         2.043 0.423 Inf   4.831  <.0001
 Female - Youth        2.270 0.424 Inf   5.358  <.0001
 Male - Youth          0.227 0.350 Inf   0.649  0.9160

P value adjustment: tukey method for comparing a family of 4 estimates
```

Among the individual interviewees (Table 5), there were also some differences in opinion. Community members tended to say there had been a significant reduction in fire frequency inside their CF, whereas NGO representatives and DFOs more often suggested a moderate reduction or no change. However, these differences were not significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    fire = case_match(
      impact_on_sfm_fire_incidence_reduction_cfma,
      "no_incidents_have_increased" ~ "Increase",
      "no_change"     ~ "No change",
      "yes_moderate_reduction"    ~ "Moderate",
      "yes_significant_reduction" ~ "Significant",
      "No change observed as at now" ~ "No change",
      .default = NA
    ),
    fire = factor(fire,
                  levels = c("Significant", "Moderate", "No change", "Increase"),
                  ordered = TRUE),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
```

```r
  ) |>
  filter(!is.na(fire)) |>
  count(interviewee_role, fire, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, fire, prop) |>
  pivot_wider(names_from = fire, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(Significant, Moderate, `No change`, Increase), decimals = 0) |>
  tab_header(
    title    = "Perceived reduction in fire incidence inside the community forest by intervie
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 5: Perceived reduction in fire incidence inside the community forest by individual interviewee role.

Perceived reduction in fire incidence inside the community forest by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Significant	Moderate	No change	Increase
CFMG executive member	63	73%	27%	0%	0%
Community member	30	67%	27%	3%	3%
Traditional leader	75	63%	24%	4%	9%
District Forestry Officer	54	44%	39%	17%	0%
NGO representative	31	45%	45%	3%	6%

Fire incidence - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had reduced fire incidence, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    fire = case_match(
      impact_on_sfm_fire_incidence_reduction_cfma,
      "no_incidents_have_increased" ~ "Increase",
      "no_change"     ~ "No change",
      "yes_moderate_reduction"    ~ "Moderate",
      "yes_significant_reduction" ~ "Significant",
      "No change observed as at now" ~ "No change",
      .default = NA
    ),
    fire = factor(fire,
                  levels = c("Increase","No change", "Moderate", "Significant"),
                  ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(fire) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$fire) >= 2) {
  M10 <- clmm(fire ~
                interviewee_role +
                interviewee_gender +
                interviewee_age +
                (1 | province) +
                (1 | province_cfmg),
              data = test.data)
  emm.M10      <- emmeans(M10, specs = ~ interviewee_role)
  pairwise.M10 <- pairs(emm.M10, adjust = "Tukey")
} else {
  M10 <- NULL
```

```r
  cat("Model not fitted: only", nlevels(test.data$fire),
      "level(s) present in fire after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: fire ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  253  -228.67 479.34 645(1988) 3.44e-04 1.2e+05

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.03049  0.1746
 province      (Intercept) 0.33360  0.5776
Number of groups:  province_cfmg 74,  province 8

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.340787   0.513994   0.663    0.507
interviewee_roletrad_leader      -0.266303   0.514472  -0.518    0.605
interviewee_roledfo              -0.613300   0.519984  -1.179    0.238
interviewee_roleNGO_rep          -0.668675   0.561838  -1.190    0.234
interviewee_gendermale           -0.608836   0.381346  -1.597    0.110
interviewee_age                   0.009922   0.011988   0.828    0.408

Threshold coefficients:
                     Estimate Std. Error z value
Increase|No change    -3.6040     0.8905  -4.047
No change|Moderate    -2.6180     0.8529  -3.070
Moderate|Significant  -0.5324     0.8186  -0.650


 contrast                       estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member  -0.3408 0.514 Inf  -0.663  0.9642
 cfmg_member - trad_leader        0.2663 0.514 Inf   0.518  0.9856
 cfmg_member - dfo                0.6133 0.520 Inf   1.179  0.7632
 cfmg_member - NGO_rep            0.6687 0.562 Inf   1.190  0.7571
 cfmg_exec_member - trad_leader   0.6071 0.433 Inf   1.402  0.6262
 cfmg_exec_member - dfo           0.9541 0.431 Inf   2.212  0.1753
```

```
 cfmg_exec_member - NGO_rep       1.0095 0.488 Inf   2.071  0.2329
 trad_leader - dfo                0.3470 0.425 Inf   0.817  0.9256
 trad_leader - NGO_rep            0.4024 0.480 Inf   0.837  0.9190
 dfo - NGO_rep                    0.0554 0.439 Inf   0.126  0.9999

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Wildlife

We asked interviewees whether CFM had led to an increase in wildlife inside their CF.

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    wildlife = case_match(
      impact_on_sfm_wildlife_increase,
      "no_reduced"              ~ "Decrease",
      "no_change"               ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      .default = NA
    ),
    wildlife = factor(wildlife,
                      levels = c("Significant", "Moderate", "No change", "Decrease")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(wildlife) |>
  count(interview_type, wildlife) |>
  group_by(interview_type) |>
  mutate(prop = n / sum(n)) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = wildlife)) +
```

```r
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = 0.01),
    name   = "Proportion of groups"
  ) +
  scale_fill_manual(
    values = c(
      "Significant" = "#2A9D8F",
      "Moderate"    = "#F0C040",
      "No change"   = "#E07B54",
      "Decrease"    = "#C0392B"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

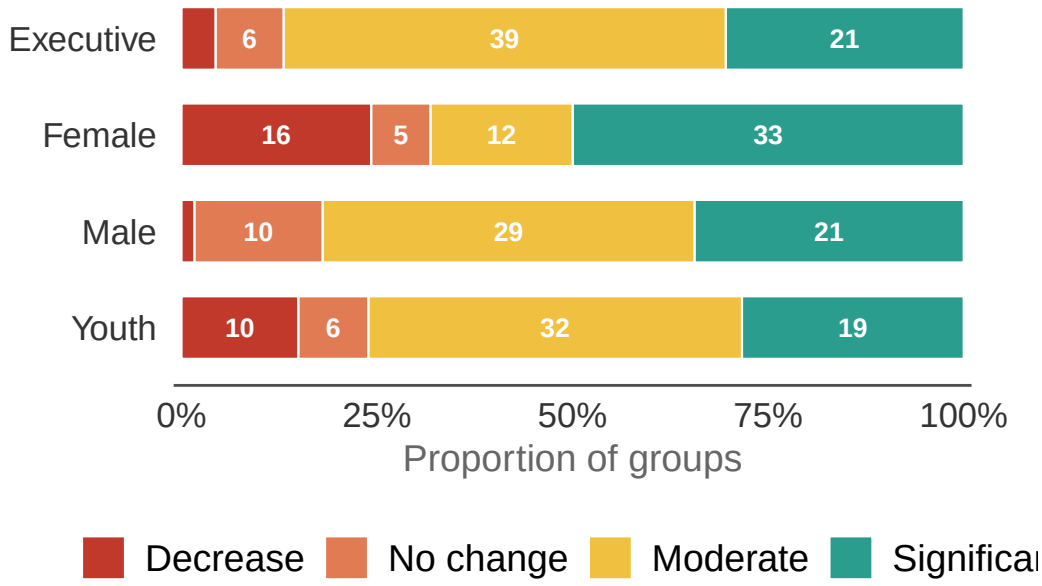

Figure 7: 'Has CFM led to an increase in wildlife in the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen, a magority of groups indicated a significant or moderate increase in wildlife in their CF. However, there were some that indicated there had been a decrease. The responses of the community female FDGs appear somewhat different from the other groups, but these differences were not significant.

> **Wildlife increase - FGDs - Model methods**
>
> To examine the differences among FGD types in their assessment of whether CFM had increased wildlife, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    wildlife = case_match(
      impact_on_sfm_wildlife_increase,
      "no_reduced"              ~ "Decrease",
      "no_change"               ~ "No change",
```

```r
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      .default = NA
    ),
    wildlife = factor(wildlife,
                      levels = c("Decrease", "No change", "Moderate", "Significant"),
                      ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(wildlife) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$wildlife) >= 2) {
  M11 <- clmm(wildlife ~
                interview_type +
                (1 | province) +
                (1 | province_cfmg),
              data = test.data)
  emm.M11      <- emmeans(M11, specs = ~ interview_type)
  pairwise.M11 <- pairs(emm.M11, adjust = "Tukey")
} else {
  M11 <- NULL
  cat("Model not fitted: only", nlevels(test.data$wildlife),
      "level(s) present in wildlife after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: wildlife ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
```

```
 logit flexible  263  -277.57 571.14 390(1851) 1.36e-04 7.9e+01

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 1.631    1.277
 province      (Intercept) 2.186    1.478
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale -0.16442    0.38126  -0.431    0.666
interview_typeMale   -0.09506    0.36884  -0.258    0.797
interview_typeYouth  -0.46440    0.35552  -1.306    0.191

Threshold coefficients:
                     Estimate Std. Error z value
Decrease|No change    -3.3585     0.6636  -5.061
No change|Moderate    -2.3430     0.6365  -3.681
Moderate|Significant   0.6224     0.6138   1.014

 contrast           estimate    SE  df z.ratio p.value
 Executive - Female   0.1644 0.381 Inf   0.431  0.9731
 Executive - Male     0.0951 0.369 Inf   0.258  0.9940
 Executive - Youth    0.4644 0.356 Inf   1.306  0.5589
 Female - Male       -0.0694 0.396 Inf  -0.175  0.9981
 Female - Youth       0.3000 0.384 Inf   0.781  0.8630
 Male - Youth         0.3693 0.373 Inf   0.989  0.7558

P value adjustment: tukey method for comparing a family of 4 estimates
```

Among the individual interviewees (Table 6), there were also some differences in opinion. For example, only 3% of NGO representatives suggested there had been a decline in wildlife in the CF, compared to between 10% and 22% for the other interviewee roles. However, again these differences were not significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    wildlife = case_match(
      impact_on_sfm_wildlife_increase,
      "no_reduced"              ~ "Decrease",
      "no_change"               ~ "No change",
```

```
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      .default = NA
    ),
    wildlife = factor(wildlife,
                      levels = c("Decrease", "No change", "Moderate", "Significant")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(wildlife)) |>
  count(interviewee_role, wildlife, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, wildlife, prop) |>
  pivot_wider(names_from = wildlife, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(Decrease, `No change`, Moderate, Significant), decimals = 0) |>
  tab_header(
    title    = "Perceived increase in wildlife by interviewee role",
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 6: Perceived increase in wildlife inside the community forest by individual interviewee role.

Perceived increase in wildlife by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Decrease	No change	Moderate	Significant
CFMG executive member	61	10%	3%	43%	44%
Community member	27	22%	15%	30%	33%
Traditional leader	73	19%	11%	34%	36%
District Forestry Officer	37	14%	24%	43%	19%
NGO representative	31	3%	10%	48%	39%

Wildlife increase - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had increased wildlife, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    wildlife = case_match(
      impact_on_sfm_wildlife_increase,
      "no_reduced"              ~ "Decrease",
      "no_change"               ~ "No change",
      "yes_moderate_increase"   ~ "Moderate",
      "yes_significant_increase" ~ "Significant",
      .default = NA
    ),
    wildlife = factor(wildlife,
                      levels = c("Decrease", "No change", "Moderate", "Significant"),
                      ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(wildlife) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
```

```r
  droplevels()

if (nlevels(test.data$wildlife) >= 2) {
  M12 <- clmm(wildlife ~
                interviewee_role +
                interviewee_gender +
                interviewee_age +
                (1 | province) +
                (1 | province_cfmg),
              data = test.data)
  emm.M12      <- emmeans(M12, specs = ~ interviewee_role)
  pairwise.M12 <- pairs(emm.M12, adjust = "Tukey")
} else {
  M12 <- NULL
  cat("Model not fitted: only", nlevels(test.data$wildlife),
      "level(s) present in wildlife after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: wildlife ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  229  -268.13 558.25 590(1773) 9.99e-05 1.1e+05

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 1.0263   1.0131
 province      (Intercept) 0.3414   0.5843
Number of groups:  province_cfmg 73,  province 8

Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.438350   0.536012   0.818   0.4135
interviewee_roletrad_leader       0.104995   0.522643   0.201   0.8408
interviewee_roledfo              -0.498814   0.562580  -0.887   0.3753
interviewee_roleNGO_rep           0.740840   0.586093   1.264   0.2062
interviewee_gendermale           -0.663377   0.391150  -1.696   0.0899 .
interviewee_age                   0.006542   0.012551   0.521   0.6022
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                     Estimate Std. Error z value
Decrease|No change    -2.3604     0.8924  -2.645
No change|Moderate    -1.4630     0.8813  -1.660
Moderate|Significant   0.8289     0.8760   0.946


 contrast                        estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member   -0.438 0.536 Inf  -0.818  0.9253
 cfmg_member - trad_leader        -0.105 0.523 Inf  -0.201  0.9996
 cfmg_member - dfo                 0.499 0.563 Inf   0.887  0.9019
 cfmg_member - NGO_rep            -0.741 0.586 Inf  -1.264  0.7134
 cfmg_exec_member - trad_leader    0.333 0.429 Inf   0.778  0.9371
 cfmg_exec_member - dfo            0.937 0.468 Inf   2.001  0.2653
 cfmg_exec_member - NGO_rep       -0.302 0.499 Inf  -0.606  0.9742
 trad_leader - dfo                 0.604 0.483 Inf   1.251  0.7212
 trad_leader - NGO_rep            -0.636 0.511 Inf  -1.243  0.7259
 dfo - NGO_rep                    -1.240 0.496 Inf  -2.499  0.0908

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Forest resources management

We asked interviewees whether CFM had improved the management of forest resources inside their CF.

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    frm = case_match(
      impact_on_sfm_forest_resources_management,
      "no_management_has_worsened"    ~ "Management worse",
      "no_change"                     ~ "No change",
      "yes_moderate_improvement"      ~ "Moderate improvement",
      "yes_significant_improvement"   ~ "Significant improvement",
      .default = NA
    ),
    frm = factor(frm, levels = c("Significant improvement", "Moderate improvement",
                                 "No change", "Management worse")),
```

```r
  interview_type = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(frm) |>
count(interview_type, frm) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = frm)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c(
    "Significant improvement" = "#2A9D8F",
    "Moderate improvement"    = "#F0C040",
    "No change"               = "#E07B54",
    "Management worse"        = "#C0392B"
  ),
  guide = guide_legend(reverse = TRUE, nrow = 2)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
```

```
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

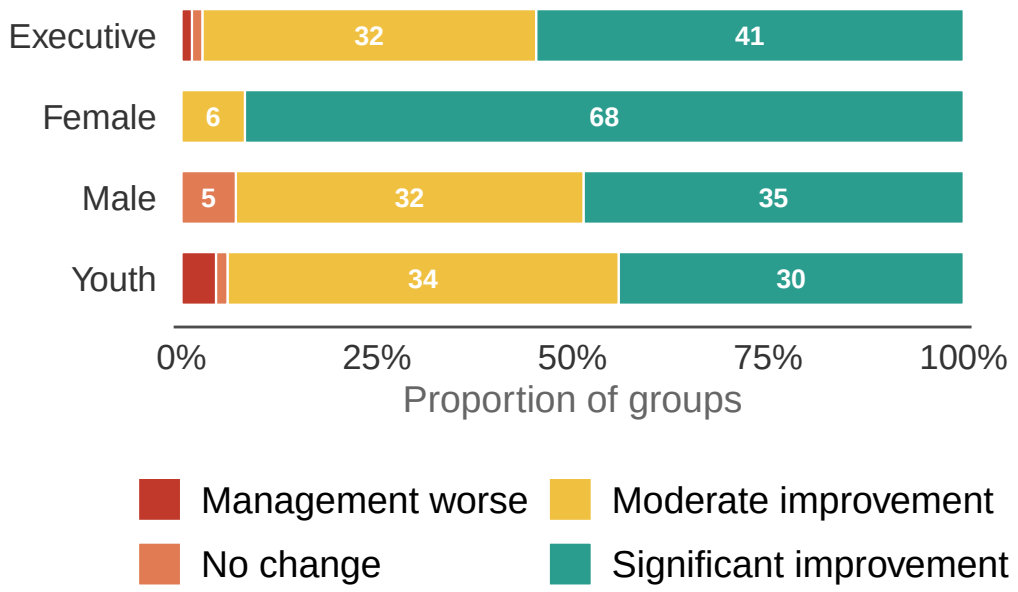

Figure 8: 'Has CFM improved forest resources management inside the community forest?' The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen, while responses vary among FGD groups, most groups agreed that the management of forest resources had improved moderately or significantly under CFM, and only a small number of primarily youth groups considered that it had gotten worse. A very high proportion of community female FDG groups considered that the management of forest resources was significantly better under CFM, a difference that was statistically significant.

Forest resources management - FGDs - Model methods

To examine the differences among FGD types in their assessment of whether CFM had improved forest resources management, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    frm = case_match(
      impact_on_sfm_forest_resources_management,
      "no_management_has_worsened"    ~ "Management worse",
      "no_change"                     ~ "No change",
      "yes_moderate_improvement"      ~ "Moderate improvement",
      "yes_significant_improvement"   ~ "Significant improvement",
      .default = NA
    ),
    frm = factor(frm,
                 levels = c("Management worse", "No change",
                            "Moderate improvement", "Significant improvement"),
                 ordered = TRUE),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(frm) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$frm) >= 2) {
  M13 <- clmm(frm ~
                interview_type +
                (1 | province) +
```

```r
                (1 | province_cfmg),
              data = test.data)
  emm.M13      <- emmeans(M13, specs = ~ interview_type)
  pairwise.M13 <- pairs(emm.M13, adjust = "Tukey")
} else {
  M13 <- NULL
  cat("Model not fitted: only", nlevels(test.data$frm),
      "level(s) present in frm after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: frm ~ interview_type + (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  289  -197.34 410.69 322(1560) 3.52e-05 2.1e+01

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 2.68e+00 1.637150
 province      (Intercept) 1.00e-12 0.000001
Number of groups:  province_cfmg 75,  province 8

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
interview_typeFemale   3.3502     0.6382   5.249 1.53e-07 ***
interview_typeMale    -0.2560     0.3862  -0.663    0.507
interview_typeYouth   -0.6343     0.3896  -1.628    0.104
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                                            Estimate Std. Error z value
Management worse|No change                   -5.6274     0.7298  -7.711
No change|Moderate improvement               -4.3720     0.5628  -7.768
Moderate improvement|Significant improvement  -0.2453     0.3374  -0.727


 contrast           estimate    SE  df z.ratio p.value
 Executive - Female   -3.350 0.638 Inf  -5.249  <.0001
 Executive - Male      0.256 0.386 Inf   0.663  0.9110
```

```
 Executive - Youth     0.634 0.390 Inf   1.628  0.3628
 Female - Male         3.606 0.632 Inf   5.707  <.0001
 Female - Youth        3.985 0.663 Inf   6.011  <.0001
 Male - Youth          0.378 0.391 Inf   0.967  0.7680

P value adjustment: tukey method for comparing a family of 4 estimates
```

We also asked which specific forest resources had improved under CFM (Figure 9).

```r
cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(
    "Pupwe"            = impact_on_sfm_harvested_forest_resources_pupwe,
    "Devil's claw"     = impact_on_sfm_harvested_forest_resources_devils_claw,
    "Fish"             = impact_on_sfm_harvested_forest_resources_fish,
    "Lusala"           = impact_on_sfm_harvested_forest_resources_lusala,
    "Mungongo"         = impact_on_sfm_harvested_forest_resourcesmungongo,
    "Bamboo"           = impact_on_sfm_harvested_forest_resources_bamboo,
    "Munkoyo"          = impact_on_sfm_harvested_forest_resources_munkoyo,
    "Poles"            = impact_on_sfm_harvested_forest_resources_poles,
    "Bushmeat"         = impact_on_sfm_harvested_forest_resources_bushmeat,
    "Timber"           = impact_on_sfm_harvested_forest_resources_timber,
    "Fibres"           = impact_on_sfm_harvested_forest_resources_fibers,
    "Chikanda"         = impact_on_sfm_harvested_forest_resources_chikanda,
    "Firewood"         = interview_grp_impact_on_sfm_harvested_forest_resources_firewood,
    "Thatching grass"  = impact_on_sfm_harvested_forest_resources_thatching_grass,
    "Medicinal plants" = impact_on_sfm_harvested_forest_resources_medicinal_plants,
    "Caterpillars"     = impact_on_sfm_harvested_forest_resources_caterpillars,
    "Honey"            = impact_on_sfm_harvested_forest_resources_honey,
    "Wild fruits"      = impact_on_sfm_harvested_forest_resources_wild_fruits,
    "Mushrooms"        = impact_on_sfm_harvested_forest_resources_mushrooms
  ) |>
  pivot_longer(everything(), names_to = "word", values_to = "value") |>
  filter(!is.na(value)) |>
  group_by(word) |>
  summarise(freq = sum(as.integer(value) == 1, na.rm = TRUE), .groups = "drop") |>
  filter(freq > 0) |>
  ggplot(aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 30)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 9: Forest resources that have improved under CFM, as reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting an improvement for each resource.

Among the individual interviewees (Table 7) the vast majority of interviewees considered that the management of forest resources had improved under CFM. There was a significant difference in responses between community members and DFOs. The DFOs were more likely to suggest there had been no change or only a moderate improvement in forest resource management under CFM, compared to the much higher proportion of interviewees indicating a significant improvement among community representatives.

```
cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    frm = case_match(
      impact_on_sfm_forest_resources_management,
```

```
      "no_management_has_worsened"    ~ "Management worse",
      "no_change"                     ~ "No change",
      "yes_moderate_improvement"      ~ "Moderate improvement",
      "yes_significant_improvement"   ~ "Significant improvement",
      .default = NA
    ),
    frm = factor(frm,
                 levels = c("Management worse", "No change",
                            "Moderate improvement", "Significant improvement")),
    interviewee_role = factor(
      case_match(
        interviewee_role,
        "cfmg_exec_member" ~ "CFMG executive member",
        "cfmg_member"      ~ "Community member",
        "trad_leader"      ~ "Traditional leader",
        "dfo"              ~ "District Forestry Officer",
        "NGO_rep"          ~ "NGO representative"
      ),
      levels = c("CFMG executive member", "Community member", "Traditional leader",
                 "District Forestry Officer", "NGO representative")
    )
  ) |>
  filter(!is.na(frm)) |>
  count(interviewee_role, frm, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, frm, prop) |>
  pivot_wider(names_from = frm, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  select(interviewee_role, n,
         `Management worse`, `No change`, `Moderate improvement`, `Significant improvement`)
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(`Management worse`, `No change`, `Moderate improvement`, `Significa
  tab_header(
    title    = "Perceived improvement in forest resources management by interviewee role",
    subtitle = "Values represent the % of respondents in each response category. n = the tota
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
```

```
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 7: Perceived change in forest resources management inside the community forest by individual interviewee role.

Perceived improvement in forest resources management by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Management worse	No change	Moderate improvement	Significant impr
CFMG executive member	63	0%	0%	29%	71%
Community member	31	3%	0%	32%	65%
Traditional leader	75	7%	5%	31%	57%
District Forestry Officer	55	4%	11%	55%	31%
NGO representative	31	0%	0%	42%	58%

Forest resources management - Individual interviewees - Model methods

To examine differences among individual interviewees in their assessment of whether CFM had improved forest resources management, we employed a cumulative link mixed model (CLMM) with random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    frm = case_match(
      impact_on_sfm_forest_resources_management,
      "no_management_has_worsened"    ~ "Management worse",
      "no_change"                     ~ "No change",
      "yes_moderate_improvement"      ~ "Moderate improvement",
      "yes_significant_improvement"   ~ "Significant improvement",
      .default = NA
    ),
    frm = factor(frm,
```

```r
                  levels = c("Management worse", "No change",
                             "Moderate improvement", "Significant improvement"),
                  ordered = TRUE),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(frm) |>
  mutate(province_cfmg = interaction(province, cfmg_name)) |>
  droplevels()

if (nlevels(test.data$frm) >= 2) {
  M14 <- clmm(frm ~
                  interviewee_role +
                  interviewee_gender +
                  interviewee_age +
                  (1 | province) +
                  (1 | province_cfmg),
                data = test.data)
  emm.M14      <- emmeans(M14, specs = ~ interviewee_role)
  pairwise.M14 <- pairs(emm.M14, adjust = "Tukey")
} else {
  M14 <- NULL
  cat("Model not fitted: only", nlevels(test.data$frm),
      "level(s) present in frm after filtering.\n")
}
```

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: frm ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
data:    test.data

 link  threshold nobs logLik  AIC    niter     max.grad cond.H
 logit flexible  255  -207.65 437.30 549(1654) 1.42e-03 2.5e+05

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.324e+00 1.1507771
 province      (Intercept) 2.080e-09 0.0000456
Number of groups:  province_cfmg 75,  province 8

Coefficients:
```

```
                                Estimate Std. Error z value Pr(>|z|)
interviewee_rolecfmg_exec_member  0.07485    0.59469   0.126  0.89984
interviewee_roletrad_leader      -0.49094    0.56077  -0.875  0.38132
interviewee_roledfo              -1.90219    0.59322  -3.207  0.00134 **
interviewee_roleNGO_rep          -0.69533    0.65299  -1.065  0.28695
interviewee_gendermale           -2.33262    0.53810  -4.335 1.46e-05 ***
interviewee_age                  -0.00916    0.01396  -0.656  0.51178
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
                                            Estimate Std. Error z value
Management worse|No change                   -7.4041     1.1664  -6.348
No change|Moderate improvement               -6.4192     1.1114  -5.776
Moderate improvement|Significant improvement  -3.2571     0.9934  -3.279


 contrast                       estimate    SE  df z.ratio p.value
 cfmg_member - cfmg_exec_member  -0.0749 0.595 Inf  -0.126  0.9999
 cfmg_member - trad_leader        0.4909 0.561 Inf   0.875  0.9060
 cfmg_member - dfo                1.9022 0.593 Inf   3.207  0.0117
 cfmg_member - NGO_rep            0.6953 0.653 Inf   1.065  0.8246
 cfmg_exec_member - trad_leader   0.5658 0.473 Inf   1.197  0.7534
 cfmg_exec_member - dfo           1.9770 0.484 Inf   4.088  0.0004
 cfmg_exec_member - NGO_rep       0.7702 0.564 Inf   1.365  0.6503
 trad_leader - dfo                1.4112 0.476 Inf   2.967  0.0250
 trad_leader - NGO_rep            0.2044 0.563 Inf   0.363  0.9963
 dfo - NGO_rep                   -1.2069 0.515 Inf  -2.344  0.1311

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
```

Restoration

We asked CFMG executives whether natural forest restoration is practiced inside the community forest. We restrict the responses for this question to CFMG Executives, because this is the groups that should be best informed about the activities of the CFMG (and the differences in responses among groups to this question are no of substantive interest).

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
```

```
  response = substr(impact_on_sfm_natural_forest_restoration, 1, 1),
  response = case_match(
    response,
    "y" ~ "Yes",
    "n" ~ "No"
  )
) |>
filter(!is.na(response)) |>
count(province, response) |>
mutate(
  n_signed = if_else(response == "No", -n, n),
  response  = factor(response, levels = c("No", "Yes"))
) |>
ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name   = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide  = guide_legend(reverse = FALSE)
) +
labs(
  y    = NULL,
  fill = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
```

```
    axis.title.x        = element_text(size = rel(1), colour = "grey40"),
    legend.position     = "bottom",
    legend.text         = element_text(size = rel(1)),
    plot.margin         = margin(10, 25, 10, 10)
  )
```

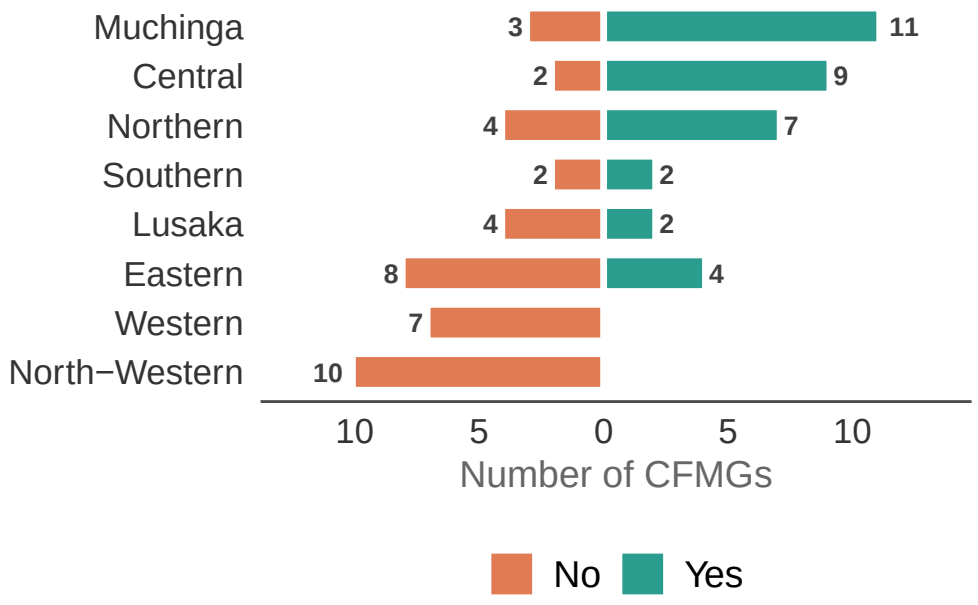

Figure 10: Whether CFMGs undertake natural forest restoration in the community forest by province. The length of the bars represents the number of CFMGs either undertaking restoration (Yes) or not (No). The numbers on the bars represent the actual number of CFMGs in each category.

Natural forest restoration is being practiced across approximately half of the CFs in our sample, mostly in Muchinga, Central and Northern provinces. We also asked CFMG executives about the specific restoration methods applied inside their CF. The majority were conserving natural regeneration. Only one CFMG was involved in planting native species and removal of invasives was not practiced.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    "Protecting natural regeneration" = impact_on_sfm_restoration_methods_applied_protection
    "Planting indigenous"             = impact_on_sfm_restoration_methods_applied_planting_w
```

```r
  "Invasives removal"               = impact_on_sfm_restoration_methods_applied_removal
) |>
pivot_longer(everything(), names_to = "method", values_to = "value") |>
filter(!is.na(value)) |>
mutate(
  response = case_match(as.character(value), "1" ~ "Yes", "0" ~ "No"),
  response = factor(response, levels = c("No", "Yes")),
  method   = factor(method, levels = c("Invasives removal",
                                       "Planting indigenous",
                                       "Protecting natural regeneration"))
) |>
count(method, response) |>
mutate(n_signed = if_else(response == "No", -n, n)) |>
ggplot(aes(y = method, x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name   = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide  = guide_legend(reverse = FALSE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
```

```
    plot.margin        = margin(10, 25, 10, 10)
  )
```

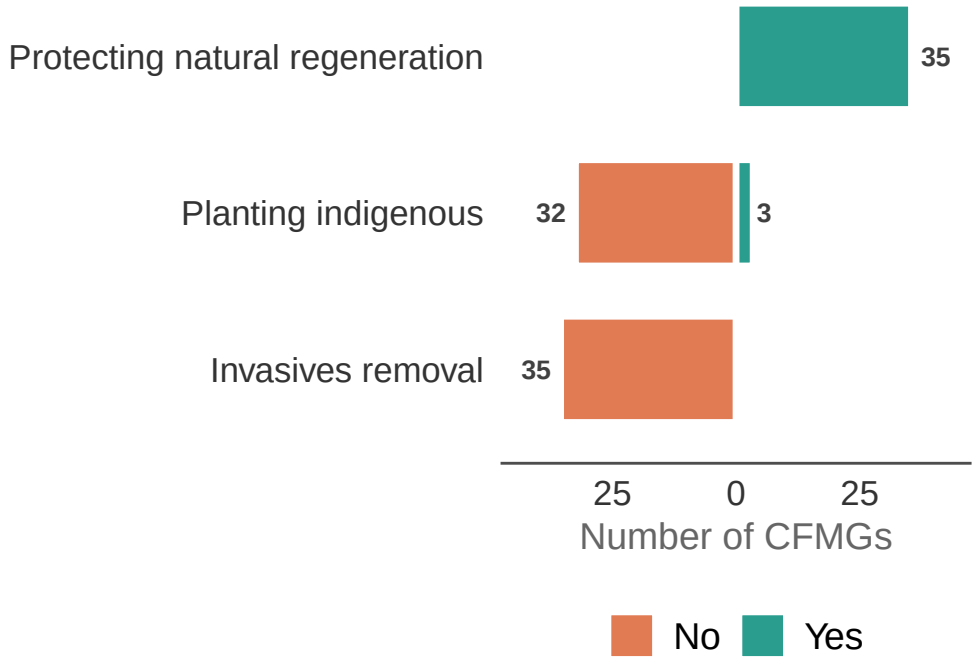

Figure 11: Restoration methods applied inside the community forest, as reported by CFMG executives. The length of the bars represents the number of CFMGs either using (Yes) or not using (No) each method. The numbers on the bars represent the actual number of CFMGs in each category.

Plantations

We also asked CFMG executives whether any plantations has been established inside the community forest.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    response = substr(impact_on_sfm_planted_forest_establishment, 1, 1),
    response = case_match(
      response,
```

```r
    "y" ~ "Yes",
    "n" ~ "No"
  )
) |>
filter(!is.na(response)) |>
count(province, response) |>
mutate(
  n_signed = if_else(response == "No", -n, n),
  response  = factor(response, levels = c("No", "Yes"))
) |>
ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name   = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide  = guide_legend(reverse = FALSE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

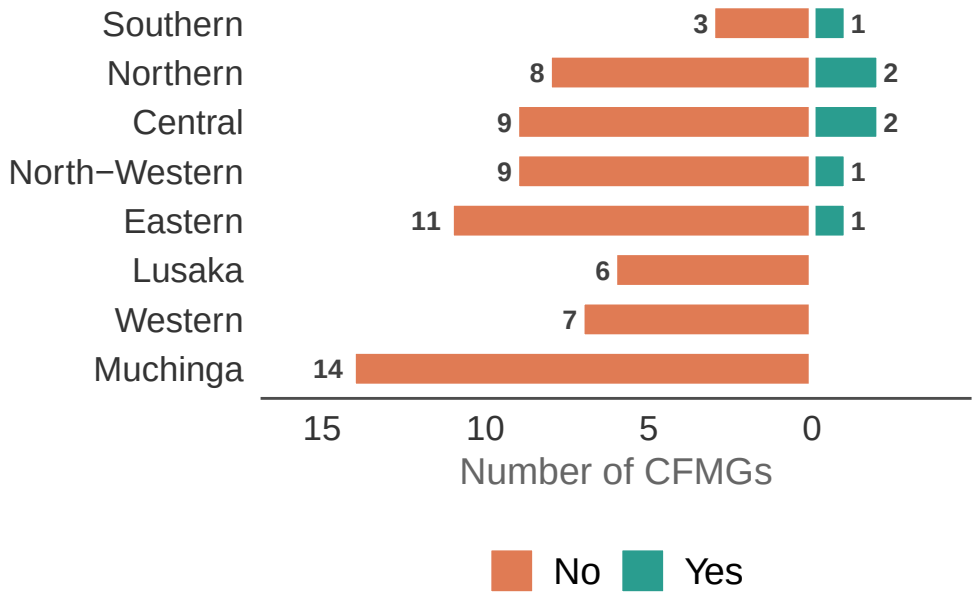

Figure 12: Whether CFMGs have established a plantation inside the community forest, by province. The length of the bars represents the number of CFMGs either with (Yes) or without (No) a plantation. The numbers on the bars represent the actual number of CFMGs in each category.

We also asked executives to estimate the number of hectares planted in the plantation. A small number (~9% of the sample) had established tree plantation inside their CF. However, the hectares planted were small (Figure 13).

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  filter(!is.na(impact_on_sfm_hectares_planted)) |>
  ggplot(aes(x = "", y = impact_on_sfm_hectares_planted)) +
  geom_boxplot(width = 0.4, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.6) +
  stat_summary(fun = mean, geom = "point", size = 3, col = "red") +
  theme_bw(base_size = 14) +
  theme(
    legend.position = "none",
    axis.text.x  = element_text(size = rel(1.14)),
    axis.text.y  = element_text(size = rel(1.14)),
    axis.title.y = element_text(size = rel(1.29))
  ) +
```

```
  xlab("") +
  ylab("Hectares planted")
```

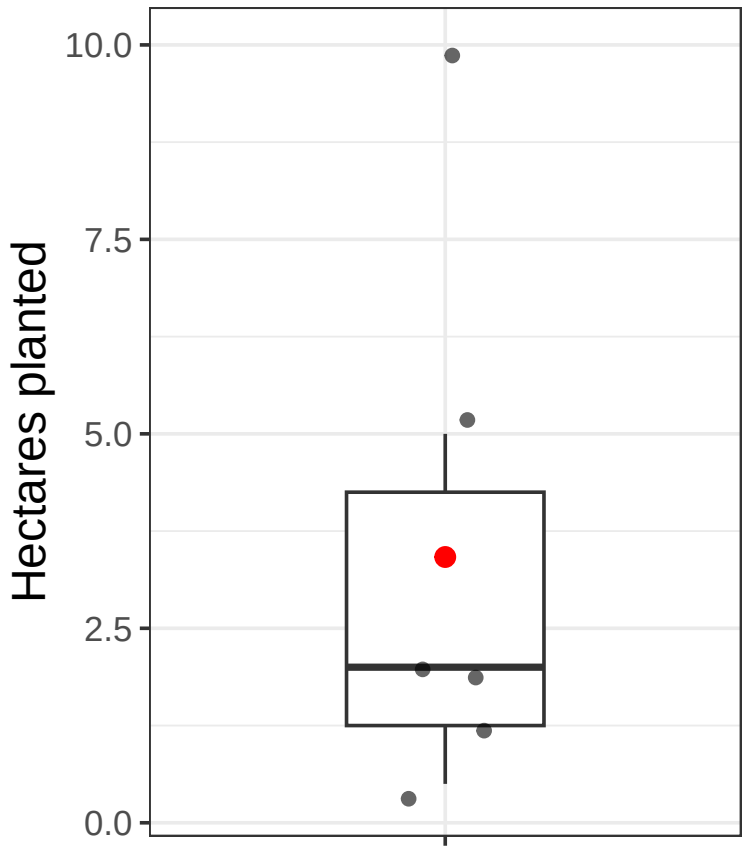

Figure 13: Estimated area of plantation established inside the community forest, as reported by CFMG executives (n = 6 CFMGs; total area = 20.5 ha). The horizontal bar is the median and the red dot is the mean. Species planted include *Gliricidia sepium* (n = 3 CFMGs), *Pinus* spp. (n = 2), *Acacia* spp. (n = 1), *Faidherbia albidia* (n = 1), and miombo species (n = 1).

Agroforestry

Last, we asked CFMG executives whether CFM has involved introducing agroforestry practices. In this context, agroforestry is implemented on agricultural land as a means for achieving

sustainable intensification and reducing pressure on forests. Around 40% of CFMGs said that agroforestry had been introduced under CFM.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    response = substr(impact_on_sfm_yn_agroforestry, 1, 1),
    response = case_match(
      response,
      "y" ~ "Yes",
      "n" ~ "No"
    )
  ) |>
  filter(!is.na(response)) |>
  count(province, response) |>
  mutate(
    n_signed = if_else(response == "No", -n, n),
    response  = factor(response, levels = c("No", "Yes"))
  ) |>
  ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label = n,
        hjust = if_else(response == "No", 1.4, -0.4)),
    size = 3.5, fontface = "bold", colour = "grey25"
  ) +
  scale_x_continuous(
    labels = ~abs(.x),
    expand = expansion(mult = 0.18),
    name   = "Number of CFMGs"
  ) +
  scale_fill_manual(
    values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
    guide  = guide_legend(reverse = FALSE)
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
```

```
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

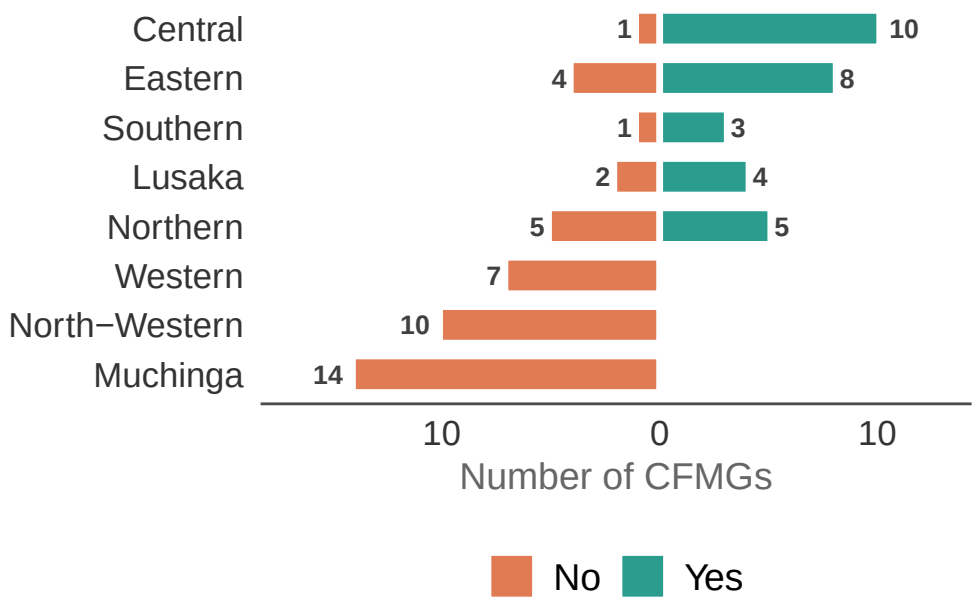

Figure 14: Whether agroforestry is employed to enhance SFM, as reported by CFMG executives. The length of the bars represents the number of CFMGs reporting either a positive impact (Yes) or no impact (No). The numbers on the bars represent the actual number of CFMGs in each category.

We also asked CFMG executives about the types of agroforestry practices present in the community.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    "Silvopasture"              = impact_on_sfm_types_agroforestry_practices_Silvopasture,
    "Windbreaks"                = impact_on_sfm_types_agroforestry_practices_wind_breaks,
    "Home gardens"              = impact_on_sfm_types_agroforestry_practices_home_gardens,
```

```r
  "Boundary planting/woodlots" = impact_on_sfm_types_agroforestry_practices_boundary_planti
  "Intercropping"              = impact_on_sfm_types_agroforestry_practices_intercropping
) |>
pivot_longer(everything(), names_to = "practice", values_to = "value") |>
filter(!is.na(value)) |>
mutate(
  response = case_match(as.character(value), "1" ~ "Yes", "0" ~ "No"),
  response = factor(response, levels = c("No", "Yes")),
  practice = factor(practice, levels = c(
    "Silvopasture", "Windbreaks", "Home gardens",
    "Boundary planting/woodlots", "Intercropping"
  ))
) |>
count(practice, response) |>
mutate(n_signed = if_else(response == "No", -n, n)) |>
ggplot(aes(y = practice, x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name   = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide  = guide_legend(reverse = FALSE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
```

```
    legend.position     = "bottom",
    legend.text         = element_text(size = rel(1)),
    plot.margin         = margin(10, 25, 10, 10)
  )
```

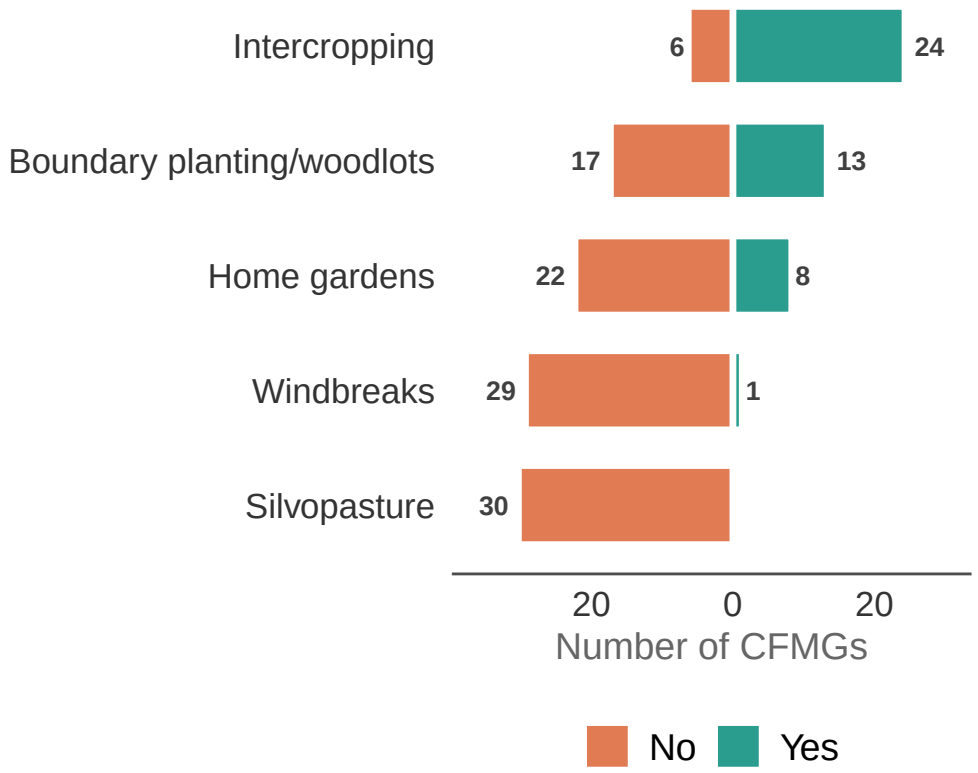

Figure 15: Types of agroforestry practices reported by CFMG executives (n = 30 CFMGs). The length of the bars represents the number of CFMGs either reporting (Yes) or not reporting (No) each practice.

We also asked CFMG executives about the species planted in agroforestry systems. Among the 30 CFMGs practicing agroforestry most were employing intercropping with leguminous shrubs for soil heal management, but planting boundary trees or woodlots and homegardens were also popular interventions.

Among the species used (Figure 16), we can see that *Gliricidia* and *Faidherbia*, leguminous shrubs and trees, respectively, were the most popular.

```r
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    "Bamboo"             = impact_on_sfm_agroforestry_species_planted_bamboo,
    "Moringa"            = impact_on_sfm_agroforestry_species_planted_moringa,
    "Exotic fruit trees" = impact_on_sfm_agroforestry_species_planted_exotic_fruit_tree,
    "Faidherbia albidia" = impact_on_sfm_agroforestry_species_planted_faidherbia_albidia,
    "Gliricidia sepium"  = impact_on_sfm_agroforestry_species_planted_gliricidia_sepium
  ) |>
  pivot_longer(everything(), names_to = "species", values_to = "value") |>
  filter(!is.na(value)) |>
  mutate(
    response = case_match(as.character(value), "1" ~ "Yes", "0" ~ "No"),
    response = factor(response, levels = c("No", "Yes")),
    species  = factor(species, levels = c(
      "Bamboo", "Moringa", "Exotic fruit trees",
      "Faidherbia albidia", "Gliricidia sepium"
    ))
  ) |>
  count(species, response) |>
  mutate(n_signed = if_else(response == "No", -n, n)) |>
  ggplot(aes(y = species, x = n_signed, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label = n,
        hjust = if_else(response == "No", 1.4, -0.4)),
    size = 3.5, fontface = "bold", colour = "grey25"
  ) +
  scale_x_continuous(
    labels = ~abs(.x),
    expand = expansion(mult = 0.18),
    name   = "Number of CFMGs"
  ) +
  scale_fill_manual(
    values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
    guide  = guide_legend(reverse = FALSE)
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
```

```
  panel.grid.minor    = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y         = element_text(size = rel(1.14), colour = "grey20",
                                     face  = "italic"),
  axis.text.x         = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x        = element_text(size = rel(1), colour = "grey40"),
  legend.position     = "bottom",
  legend.text         = element_text(size = rel(1)),
  plot.margin         = margin(10, 25, 10, 10)
)
```

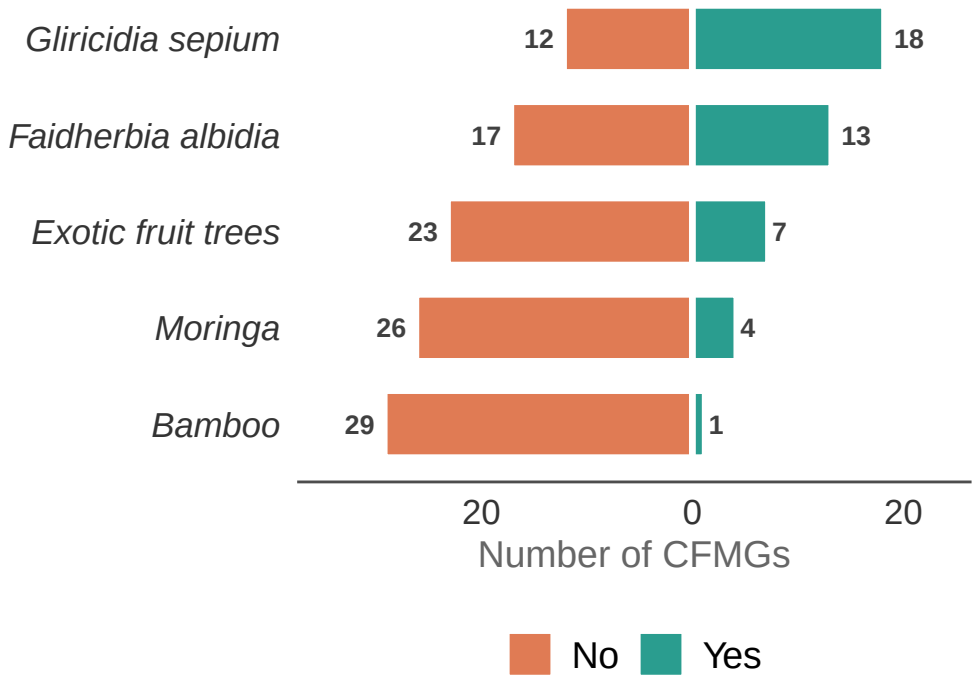

Figure 16: Agroforestry species planted, as reported by CFMG executives (n = 30 CFMGs). The length of the bars represents the number of CFMGs either planting (Yes) or not planting (No) each species.

Discussion

All the indicators of SFM point to a situation that is improving under CFM, including deforestation, charcoal production, fire frequency, wildlife abundance and natural resource management. Interestingly, there does not appear to be any indication of leakage for most CFs. Most FDG groups indicated that both deforestation and charcoal production were also decreasing outside their CF. There were some differences in responses among the different FDG groups, with community female FDGs tending to have a more positive view of the impact of CFM compared to the other groups. This is interesting and may indicate that under CFM females have more of a voice in forest and forest resource management. Traditionally, women are highly dependent on forest resources, but usually have little say in forest management decisions. It would be worth following up this study to investigate what is driving this positive take on CFM among community women.

Among the individual interviewees, NGO representatives and DFOs tended to have a somewhat more conservative view with respect to positive changes, as compared to individuals from the communities. This may reflect that these professionals have a broader perspective from which to understand changes in forest and forest resource management. However, it could also arise from the fact that they are not as intimately involved in the forest management and hence less aware of the changes. Ultimately, monitoring over a longer period will enable better understanding of how CFM is changing forest management.

Overall, CFM appears to be engendering a change towards improved SFM. However, it is important to note that a small number of FDG groups and individual interviewees suggested the situation was worsening in their CFs. For the success of CFM in Zambia, it will be important to identify CFs that are doing less well, diagnose the causes and implement changes.

Some CFMGs were practicing forest restoration, mostly through protecting natural regeneration, and a few were establishing plantations. However, these activities were not widespread among CFMGs. Restoration - especially through protection of natural regeneration - should integral to CFM and is probably occurring widely simply through improvements in forest protection. As this is an important indicator of SFM and there are government targets to restore 2,000,000 ha by 2030, the FD should integrate restoration targets into forest management plans for CFMGs and monitor outcomes. Agroforestry was also not widely practiced by CFMGs, but is recognised tool for restoration of degraded agricultural land, and can reduce pressure on forests. Plantations are also an important opportunity for CFM. Plantations can supply local building and fuelwood needs, thereby reducing pressure on natural forests. Through sustainable charcoal programmes, they can also supply energy for urban dwellers and again reduce pressure on natural forests. Last, Zambia has an acute shortage of high quality timber for telegraph poles and construction and imports large amounts of wood. A focus on developing plantations through CFM would go some way to meeting these needs, as well as generating a valuable additional source of income for poor rural communities.

In conclusion, our data suggest that CFM in Zambia is driving substantial improvements

in SFM, reducing deforestation, charcoal production, and fire frequency, increasing wildlife populations, and improving natural resource management. However, it is important to note that these changes were not ubiquitous across all CFs. While some CFMGs are practising restoration and establishing plantations, there is a substantial opportunity to expand the role of CFM in meeting national restoration and plantation targets.